



Spirent **Fit4Launch**

WFC Performance Report | **Wingtech Revvl 6X 5G**

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**T-Mobile**

|         |                             |
|---------|-----------------------------|
| DATE    | May 5, 2023                 |
| VERSION | 1.0                         |
| CONTACT | Chuck Bedrosian             |
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User Experience Analytics by Spirent

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## Device Details

| Device                                                                                                   | Role        | IMEI                                                                     | Software Version                  | Hardware Version |
|----------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------|-----------------------------------|------------------|
| Wingtech<br>Revvl 6X 5G                                                                                  | Test Device | 864947060004337<br>864947060003883<br>864947060003958<br>864947060003933 | TMRV065G_0.01.02                  | V1.0             |
| Samsung<br>SM-G981U                                                                                      | Reference   | 355611110676876<br>354268111460924                                       | TP1A.220624.014.G98<br>1USQS3HWB6 | REV1.0           |
| Test Date: 05/01/2023 – 05/04/2023                                                                       |             |                                                                          |                                   |                  |
| <b>Test Location:</b> Spirent Laboratory, Frederick MD<br><b>Network Environment:</b> Cellular and Wi-Fi |             |                                                                          |                                   |                  |

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### WiFi Stats

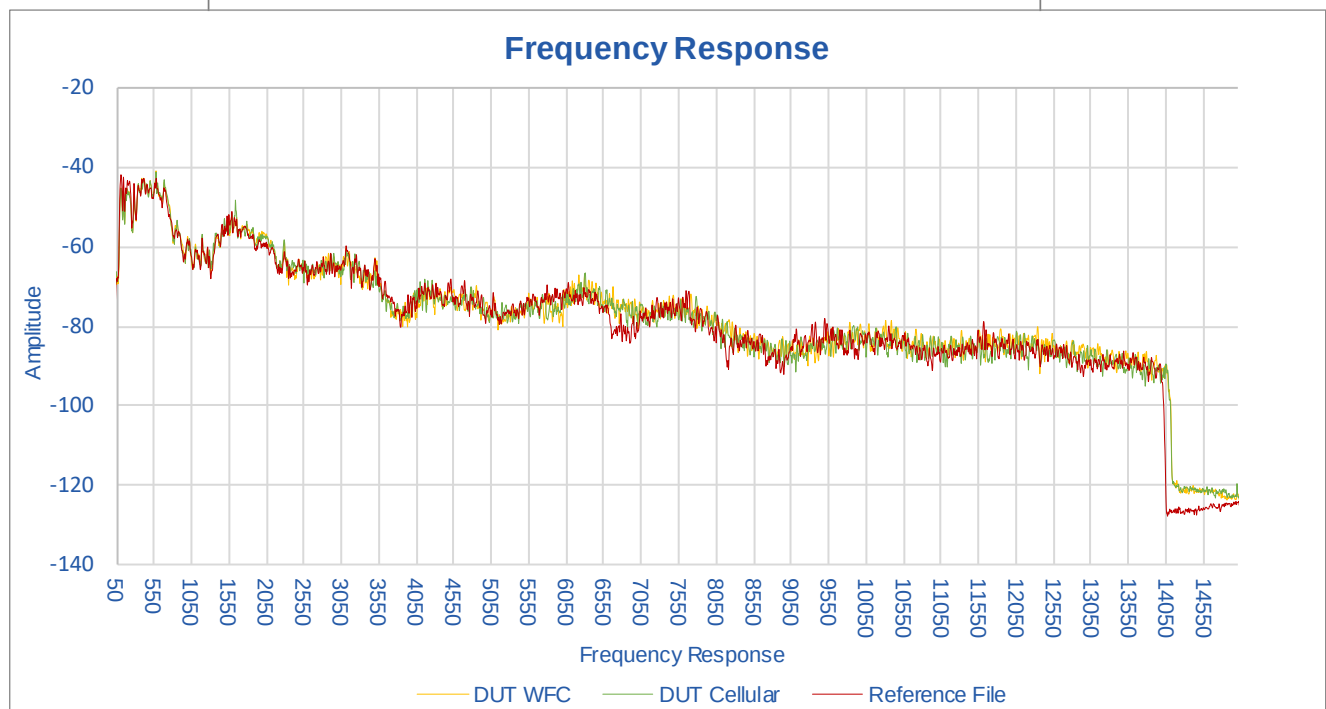
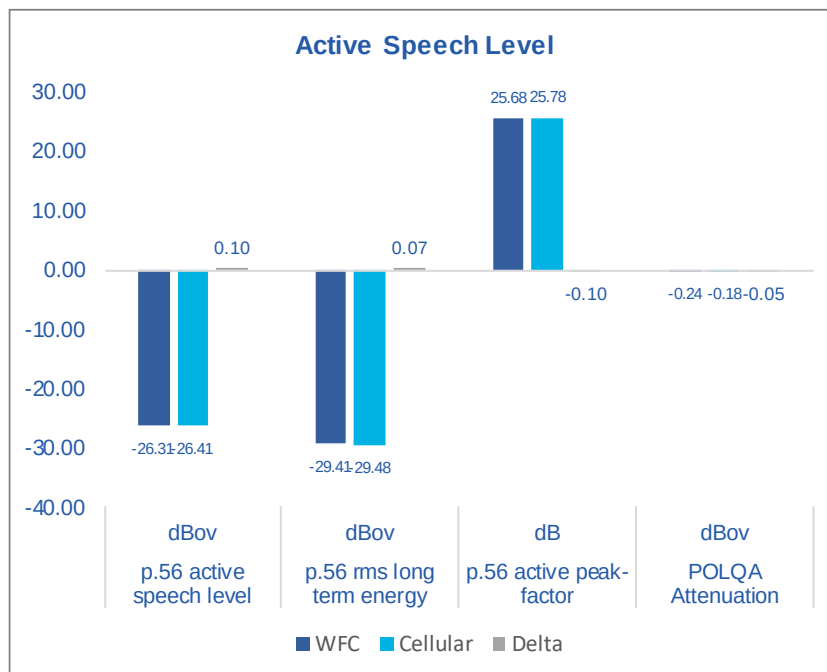
### Appendix

## Overall Call Performance

## Baseline Audio Performance

## Baseline Audio Performance

| Baseline Audio Performance |      |        |          |       |
|----------------------------|------|--------|----------|-------|
| Audio NAT Functional       | Unit | WFC    | Cellular | Delta |
| p.56 active speech level   | dBov | -26.31 | -26.41   | 0.10  |
| p.56 rms long term energy  | dBov | -29.41 | -29.48   | 0.07  |
| p.56 active peak-factor    | dB   | 25.68  | 25.78    | -0.10 |
| POLQA Attenuation          | dBov | -0.24  | -0.18    | -0.05 |



## Baseline Cellular and WFC Call Performance

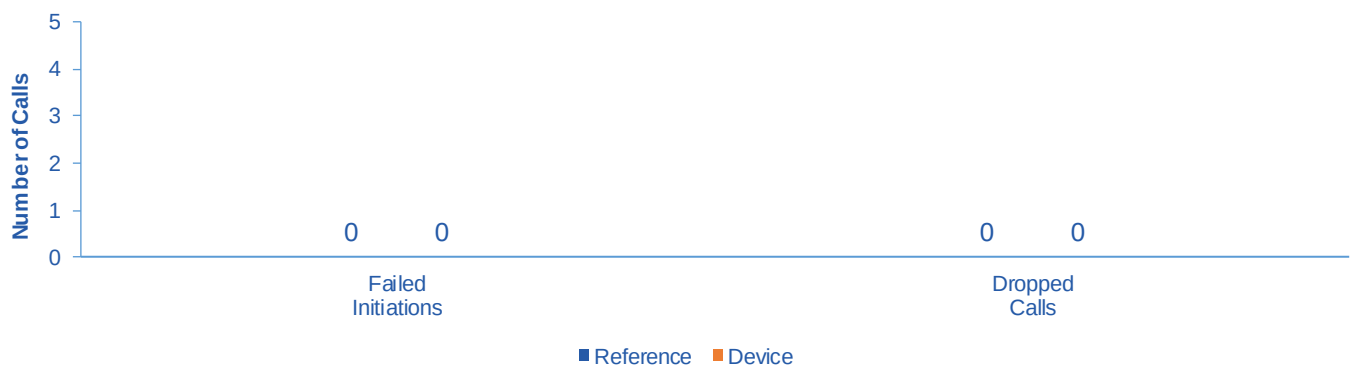


## Cellular Call Performance and Audio Quality Baseline

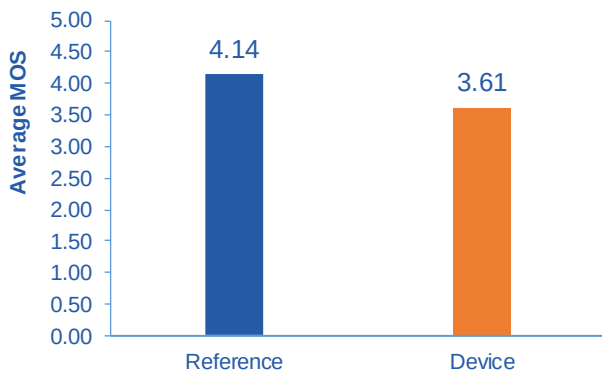
| Cellular Call Performance and Audio Quality Baseline |             |          |                     |                     |                        |                    |                  |               |      |
|------------------------------------------------------|-------------|----------|---------------------|---------------------|------------------------|--------------------|------------------|---------------|------|
| Network                                              | Device Role | RSRP dBm | Connection Attempts | Mean Setup Time (s) | Successful Initiations | Failed Initiations | Successful Calls | Dropped Calls | MOS  |
| No Impairment                                        | Reference   | -114     | 5                   | 6.83                | 100.0%                 | 0                  | 100.0%           | 0             | 4.14 |
| No Impairment                                        | Device      | -114     | 5                   | 5.37                | 100.0%                 | 0                  | 100.0%           | 0             | 3.61 |

Note: All the values are from Mobile Origination (MO)

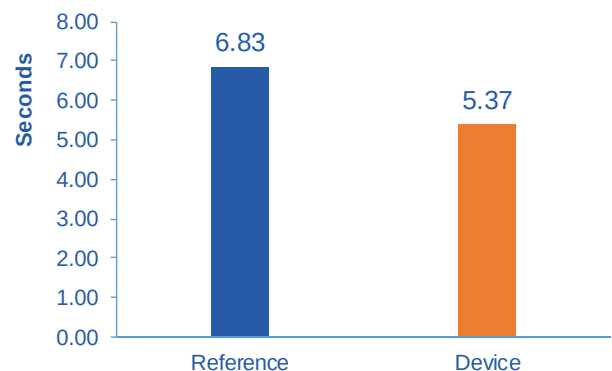
### Cellular Call Performance



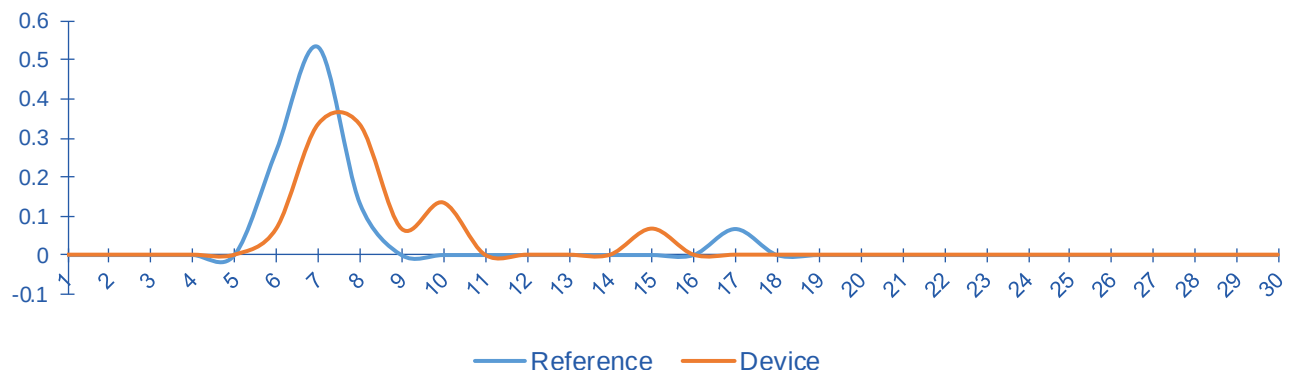
### POLQA MOS - Cellular



### Mean Setup Time - Cellular



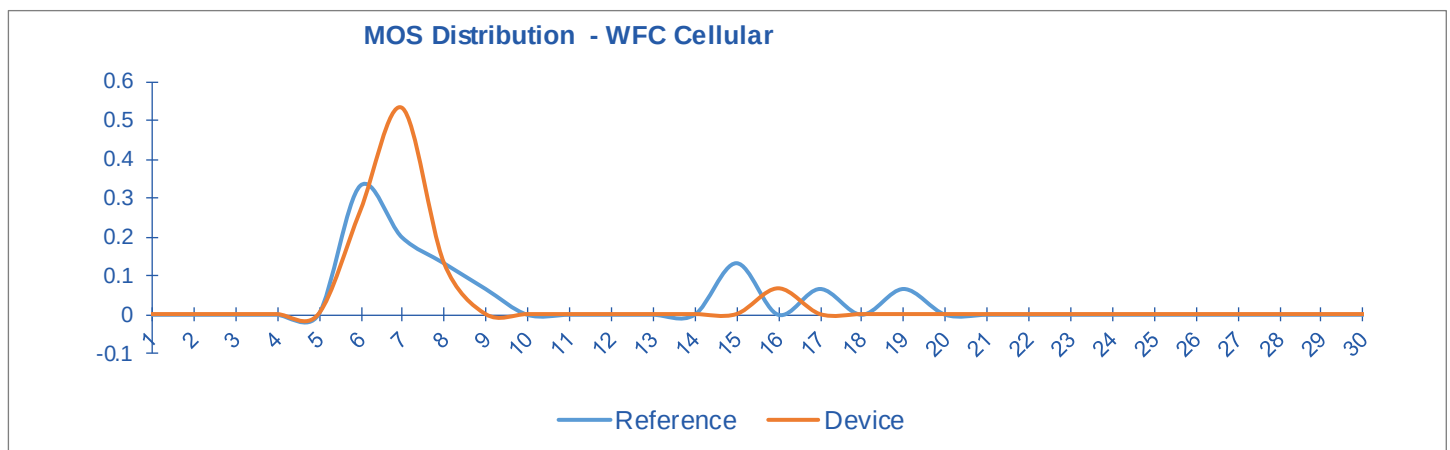
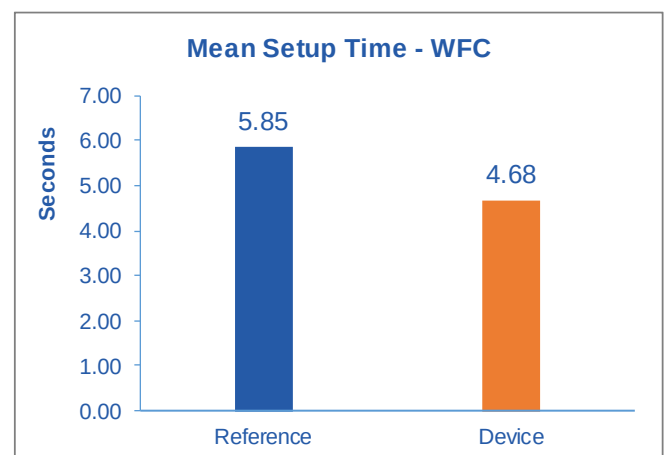
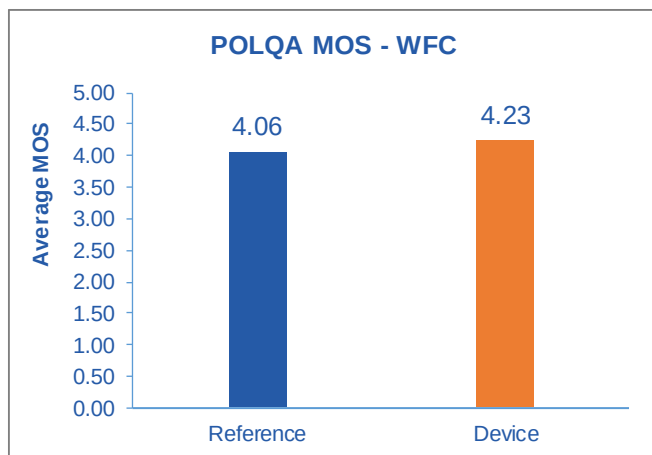
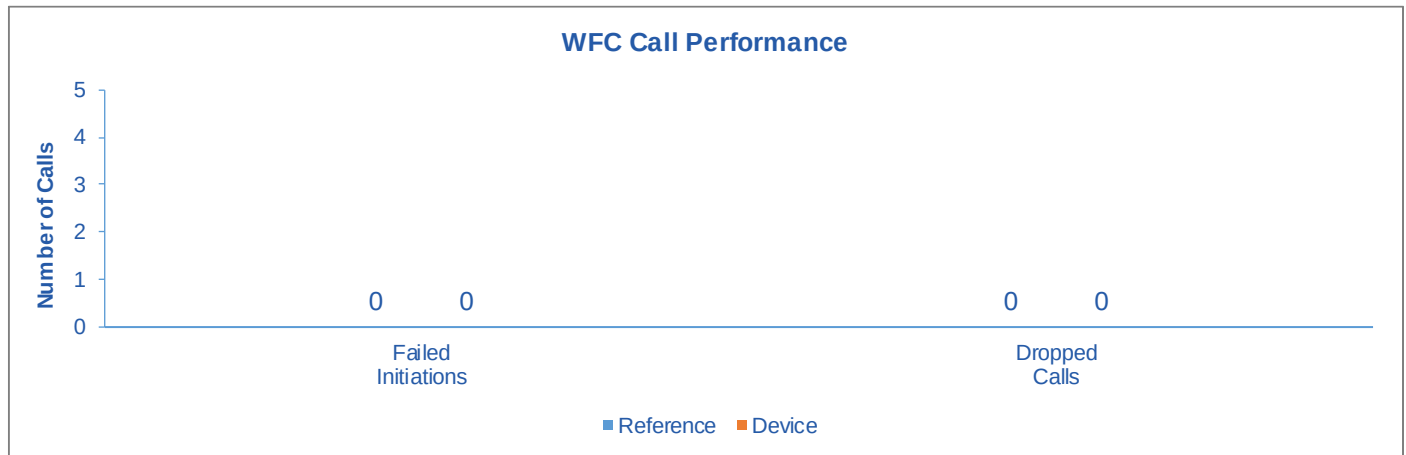
### MOS Distribution - VoLTE Cellular



## WFC Call Performance and Audio Quality Baseline (Linksys WFC)

| WFC Call Performance and Audio Quality Baseline (Linksys WFC) |             |          |                     |                     |                        |                    |                  |               |      |
|---------------------------------------------------------------|-------------|----------|---------------------|---------------------|------------------------|--------------------|------------------|---------------|------|
| Network                                                       | Device Role | RSRP dBm | Connection Attempts | Mean Setup Time (s) | Successful Initiations | Failed Initiations | Successful Calls | Dropped Calls | MOS  |
| Baseline                                                      | Reference   | -65      | 5                   | 5.85                | 100.0%                 | 0                  | 100.0%           | 0             | 4.06 |
| Baseline                                                      | Device      | -65      | 5                   | 4.68                | 100.0%                 | 0                  | 100.0%           | 0             | 4.23 |

Note: All the values are from Mobile Origination (MO)



## WFC Call Performance

## WFC Call Performance - Overall

| WFC Overall Call Performance |                     |                     |                                |               |                            |                       |
|------------------------------|---------------------|---------------------|--------------------------------|---------------|----------------------------|-----------------------|
| WFC                          | Connection Attempts | Mean Setup Time (s) | Failed Initiations Initiations | Dropped Calls | Failed Initiations P-Value | Dropped Calls P-Value |
| Reference                    | 216                 | 6.87                | 0                              | 0             | 0.50                       | 1.00                  |
| Device                       | 216                 | 5.67                | 1                              | 0             |                            |                       |

Note: All the values are from Mobile Origination (MO)

### \* Pass/Fail Criteria

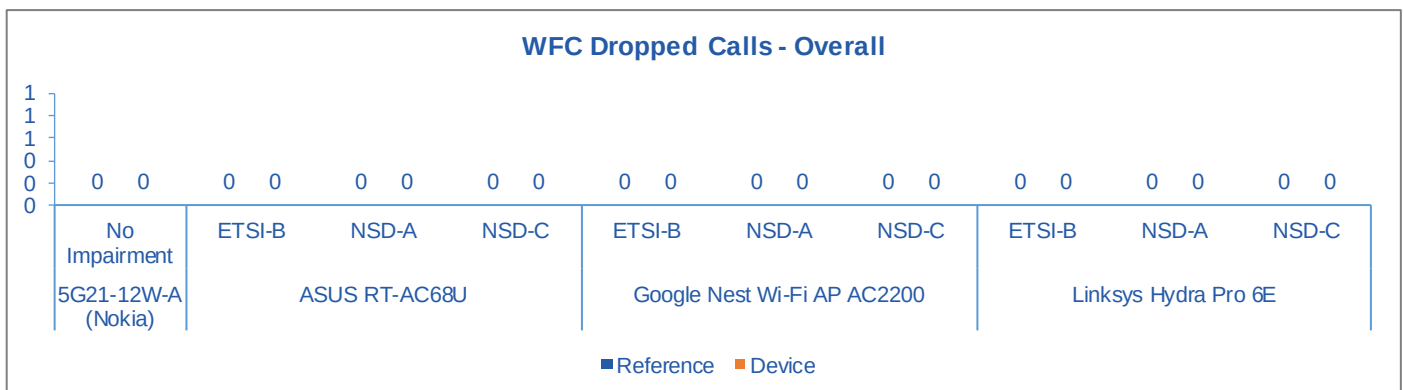
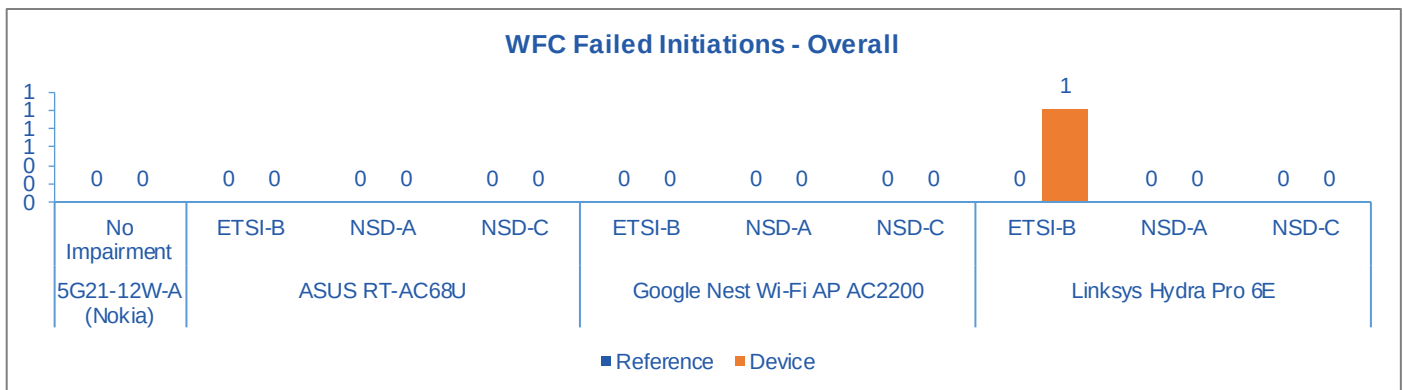
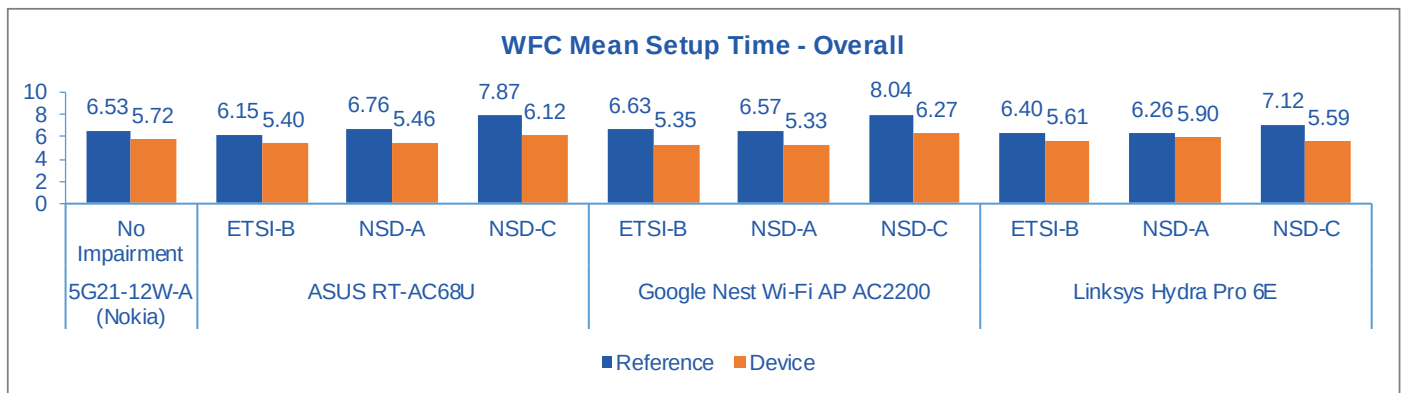
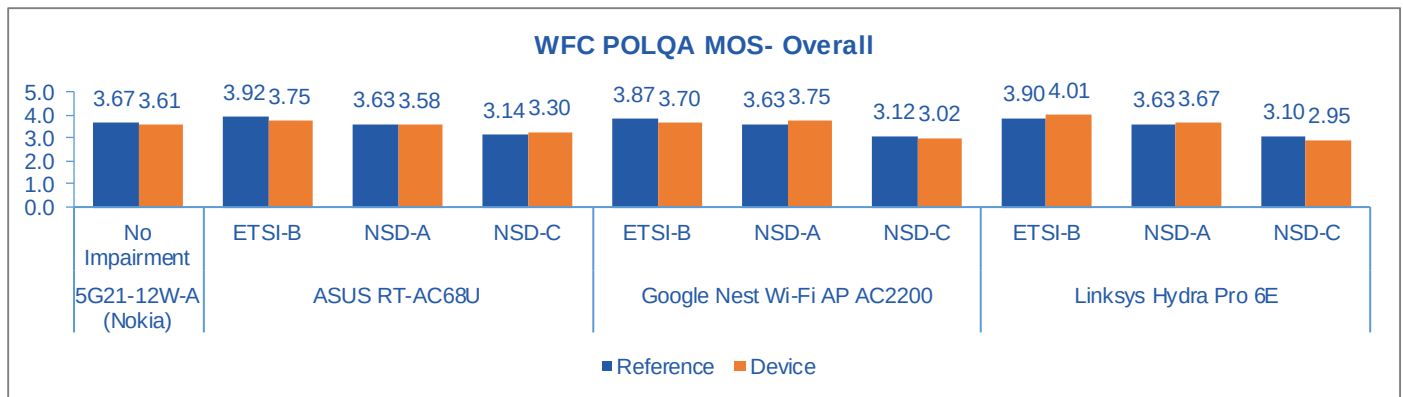
| Call Performance ETSI-A, NSD-B, NSD-C |                                                             |
|---------------------------------------|-------------------------------------------------------------|
| Mean Setup Time                       | if(Avg of DUT) < (Avg of REF)                               |
|                                       | if(Avg of DUT) < (Avg of REF*1.1)                           |
|                                       | if(Avg of DUT) between (Avg of REF)*1.25 & 1.1              |
|                                       | if(Avg of DUT) > (Avg of REF*1.25)                          |
| Success/Drops                         | If p-value > 0.95 for DUT vs REF                            |
|                                       | If p-value between 0.05 and 0.95 for DUT vs REF             |
|                                       | RED condition is satisfied but % failures/Drops of DUT < 1% |
|                                       | If p-value for Fisher Exact < 0.05 for DUT vs REF           |
| MOS                                   | If DUT > REF                                                |
|                                       | If DUT > REF-0.1                                            |
|                                       | If Red or Green Not Satisfied                               |
|                                       | If DUT<REF-0.25                                             |

## WFC Call Performance - Overall

| WFC Call Performance        |           |                     |                     |                                |               |                            |                       |
|-----------------------------|-----------|---------------------|---------------------|--------------------------------|---------------|----------------------------|-----------------------|
| WFC                         |           | Connection Attempts | Mean Setup Time (s) | Failed Initiations Initiations | Dropped Calls | Failed Initiations P-Value | Dropped Calls P-Value |
| ASUS RT-AC68U               | Reference | 72                  | 6.93                | 0                              | 0             | 1.00                       | 1.00                  |
|                             | Device    | 72                  | 5.66                | 0                              | 0             |                            |                       |
| Google Nest Wi-Fi AP AC2200 | Reference | 72                  | 7.08                | 0                              | 0             | 1.00                       | 1.00                  |
|                             | Device    | 72                  | 5.65                | 0                              | 0             |                            |                       |
| Linksys Hydra Pro 6E        | Reference | 72                  | 6.60                | 0                              | 0             | 0.50                       | 1.00                  |
|                             | Device    | 72                  | 5.70                | 1                              | 0             |                            |                       |

| WFC Overall AudioPerformance |           |                     |        |       |       |
|------------------------------|-----------|---------------------|--------|-------|-------|
| MOS                          |           | Connection Attempts | ETSI-B | NSD-A | NSD-C |
| ASUS RT-AC68U                | Reference | 72                  | 3.92   | 3.63  | 3.14  |
|                              | Device    | 72                  | 3.75   | 3.58  | 3.30  |
| Google Nest Wi-Fi AP AC2200  | Reference | 72                  | 3.87   | 3.63  | 3.12  |
|                              | Device    | 72                  | 3.70   | 3.75  | 3.02  |
| Linksys Hydra Pro 6E         | Reference | 72                  | 3.90   | 3.63  | 3.10  |
|                              | Device    | 72                  | 4.01   | 3.67  | 2.95  |

## WFC Call Performance – Overall

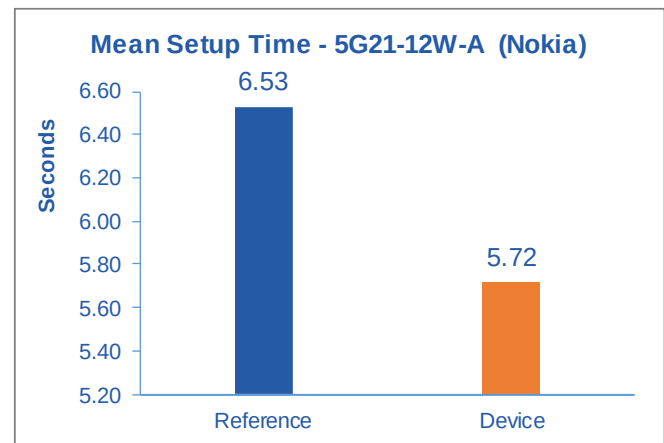
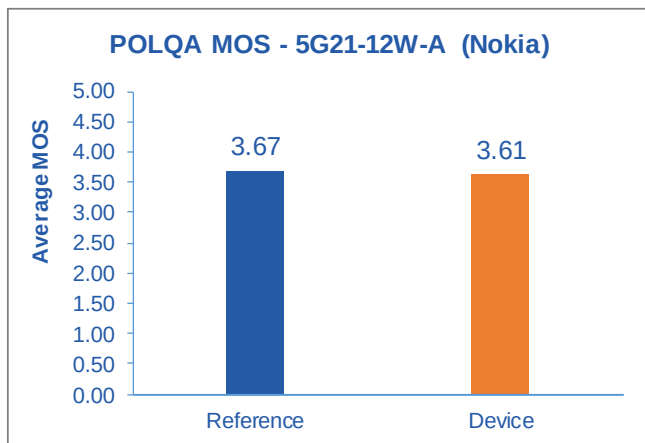
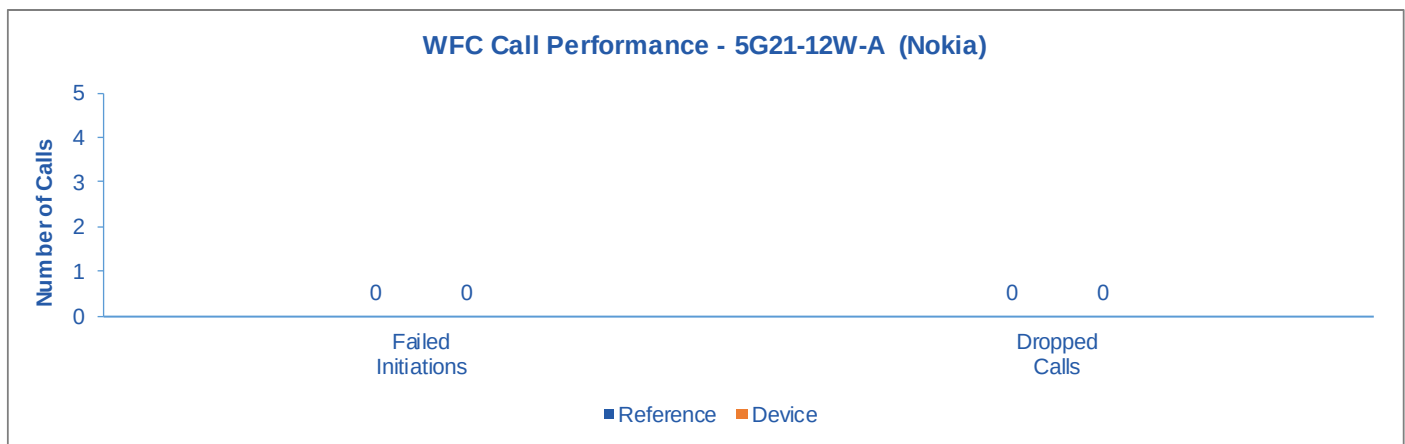


## WFC Call Performance

### 5G21-12W-A (Nokia)

| WFC Call Performance - 5G21-12W-A (Nokia) |             |          |                     |                     |                        |                    |                  |               |      |
|-------------------------------------------|-------------|----------|---------------------|---------------------|------------------------|--------------------|------------------|---------------|------|
| WFC                                       | Device Role | RSSI dBm | Connection Attempts | Mean Setup Time (s) | Successful Initiations | Failed Initiations | Successful Calls | Dropped Calls | MOS  |
| No Impairment                             | Reference   | -65      | 24                  | 6.53                | 100.0%                 | 0                  | 100.0%           | 0             | 3.67 |
| No Impairment                             | Device      | -65      | 24                  | 5.72                | 100.0%                 | 0                  | 100.0%           | 0             | 3.61 |

Note: All the values are from Mobile Origination (MO)

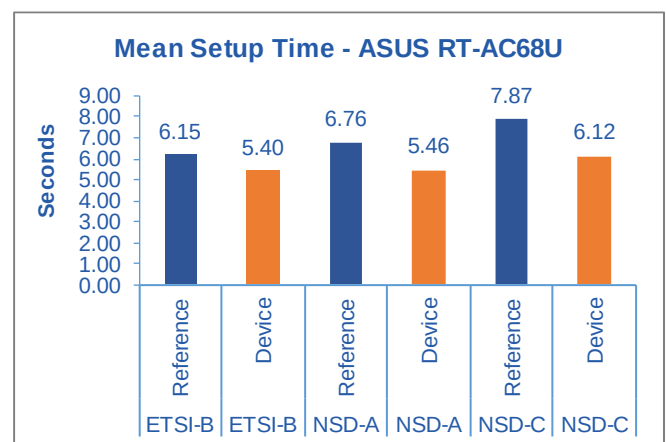
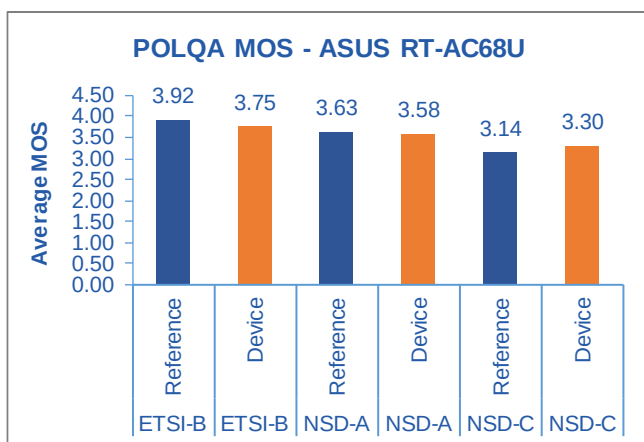
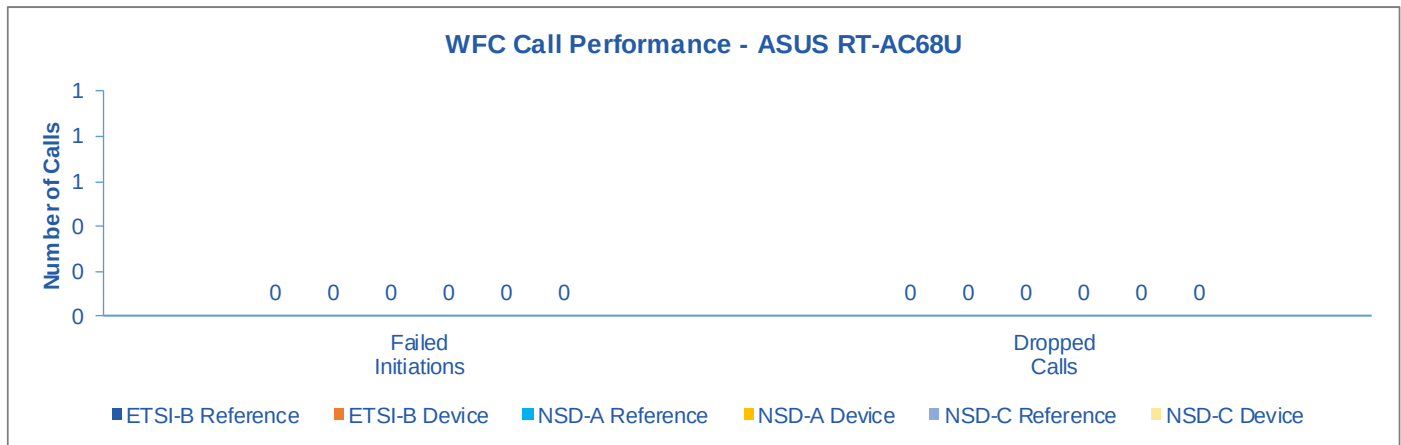


## WFC Call Performance

### ASUS RT-AC68U

| WFC Call Performance - ASUS RT-AC68U |             |          |                     |                     |                        |                    |                  |               |      |
|--------------------------------------|-------------|----------|---------------------|---------------------|------------------------|--------------------|------------------|---------------|------|
| WFC                                  | Device Role | RSSI dBm | Connection Attempts | Mean Setup Time (s) | Successful Initiations | Failed Initiations | Successful Calls | Dropped Calls | MOS  |
| ETSI-B                               | Reference   | -65      | 24                  | 6.15                | 100.0%                 | 0                  | 100.0%           | 0             | 3.92 |
| ETSI-B                               | Device      | -65      | 24                  | 5.40                | 100.0%                 | 0                  | 100.0%           | 0             | 3.75 |
| NSD-A                                | Reference   | -65      | 24                  | 6.76                | 100.0%                 | 0                  | 100.0%           | 0             | 3.63 |
| NSD-A                                | Device      | -65      | 24                  | 5.46                | 100.0%                 | 0                  | 100.0%           | 0             | 3.58 |
| NSD-C                                | Reference   | -65      | 24                  | 7.87                | 100.0%                 | 0                  | 100.0%           | 0             | 3.14 |
| NSD-C                                | Device      | -65      | 24                  | 6.12                | 100.0%                 | 0                  | 100.0%           | 0             | 3.30 |

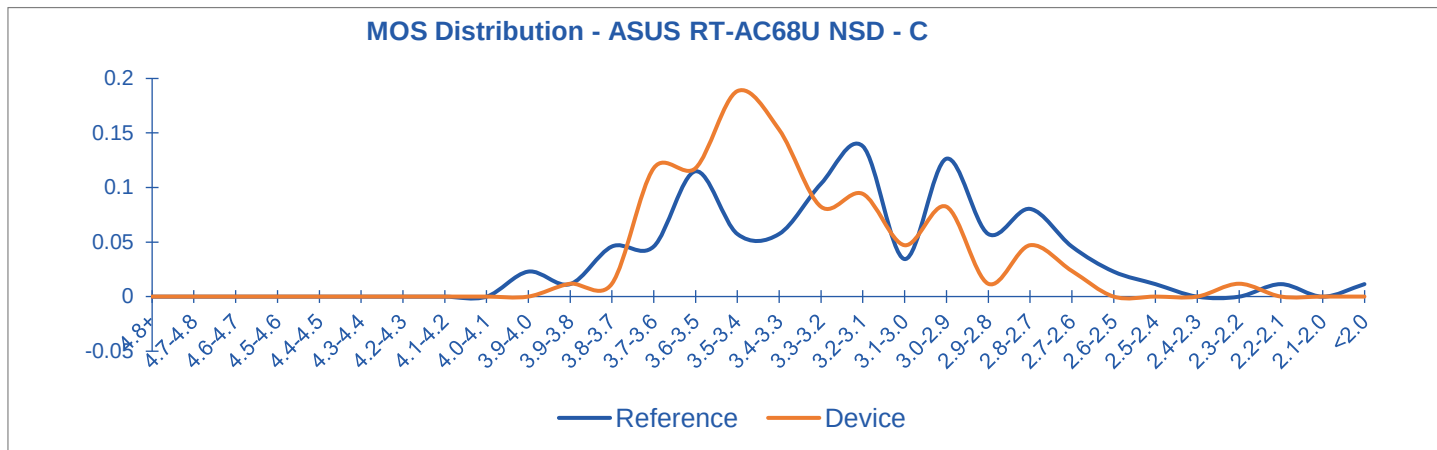
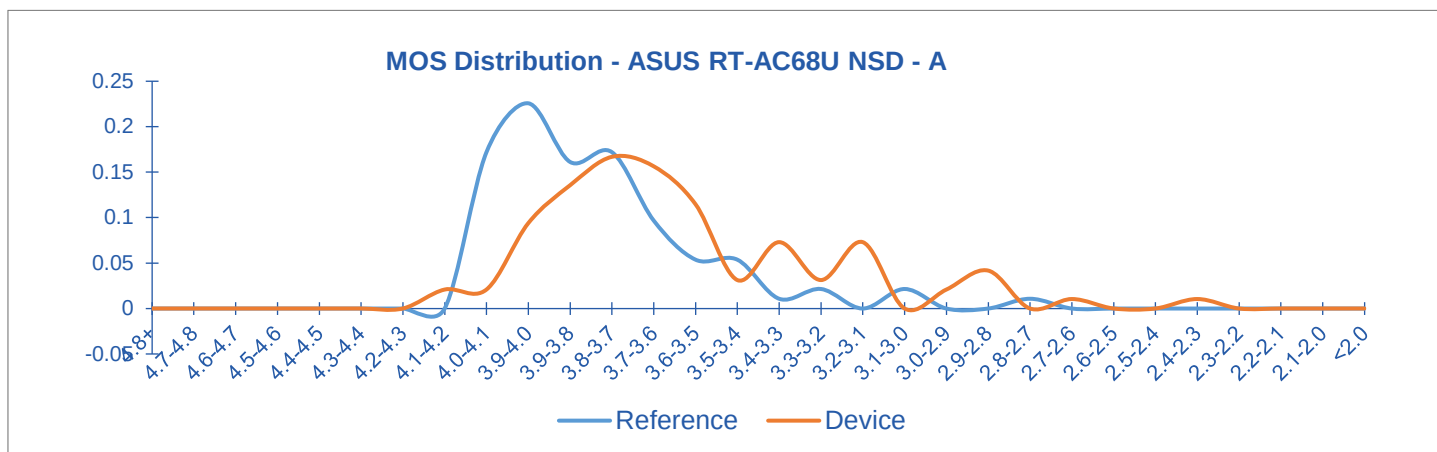
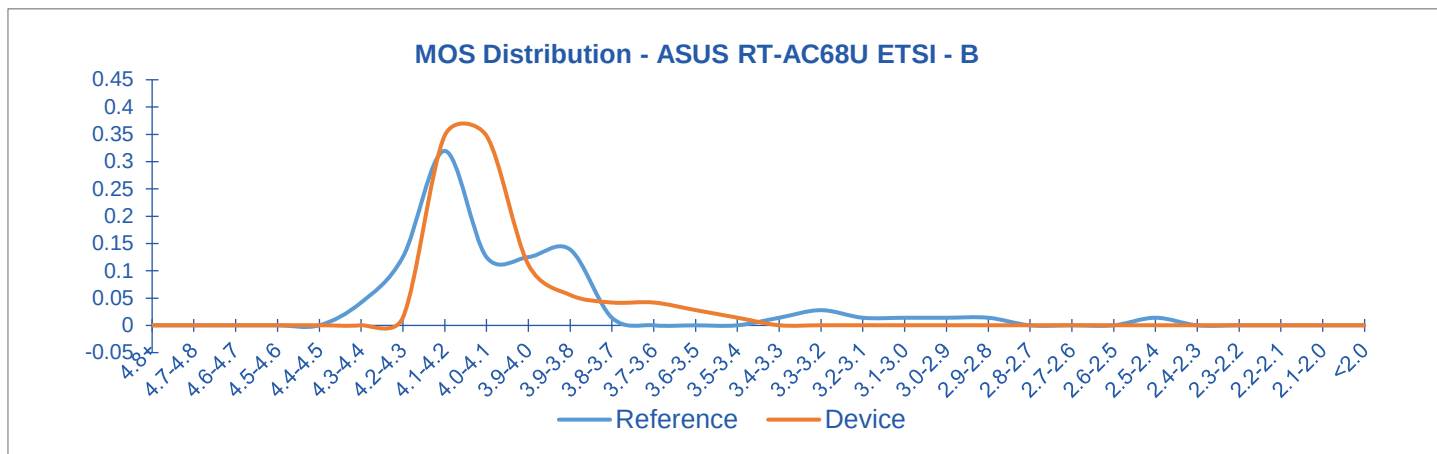
Note: All the values are from Mobile Origination (MO)





## WFC Call Performance

### ASUS RT-AC68U

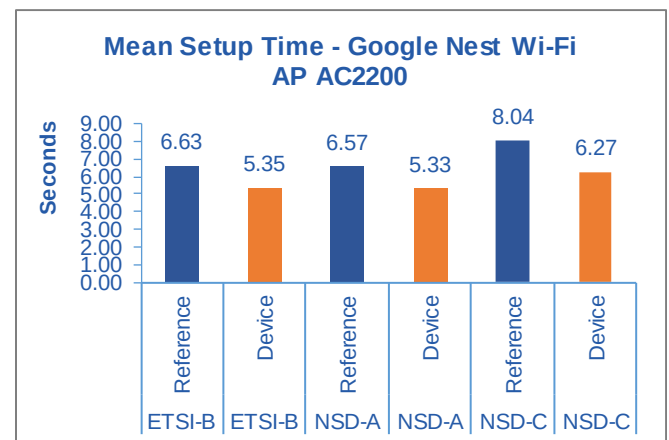
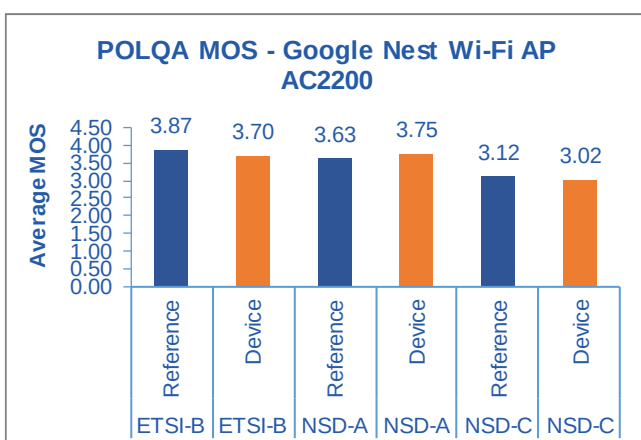
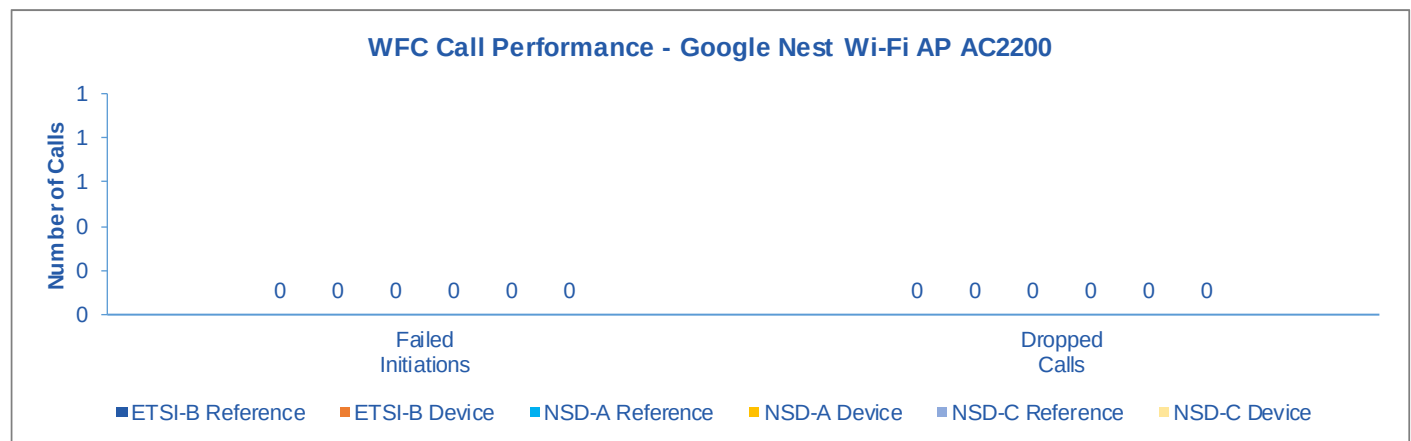


## WFC Call Performance

### Google Nest Wi-Fi AP AC2200

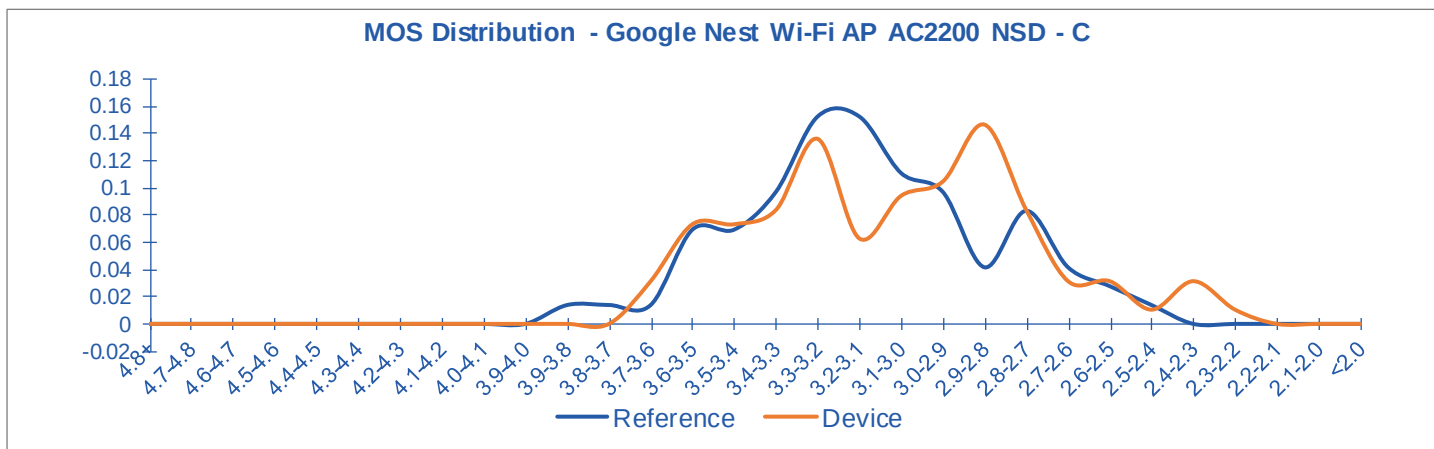
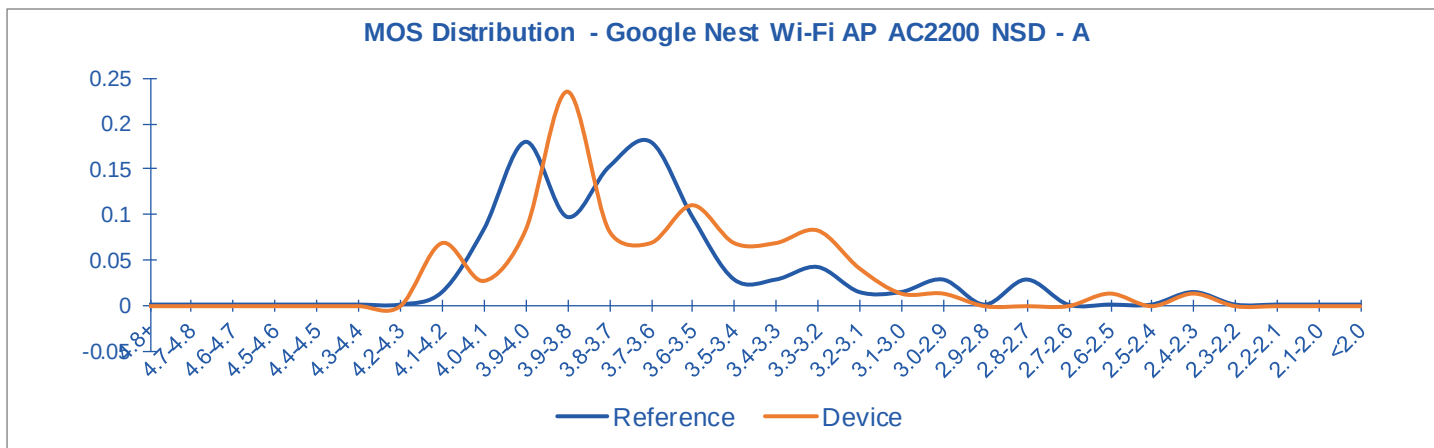
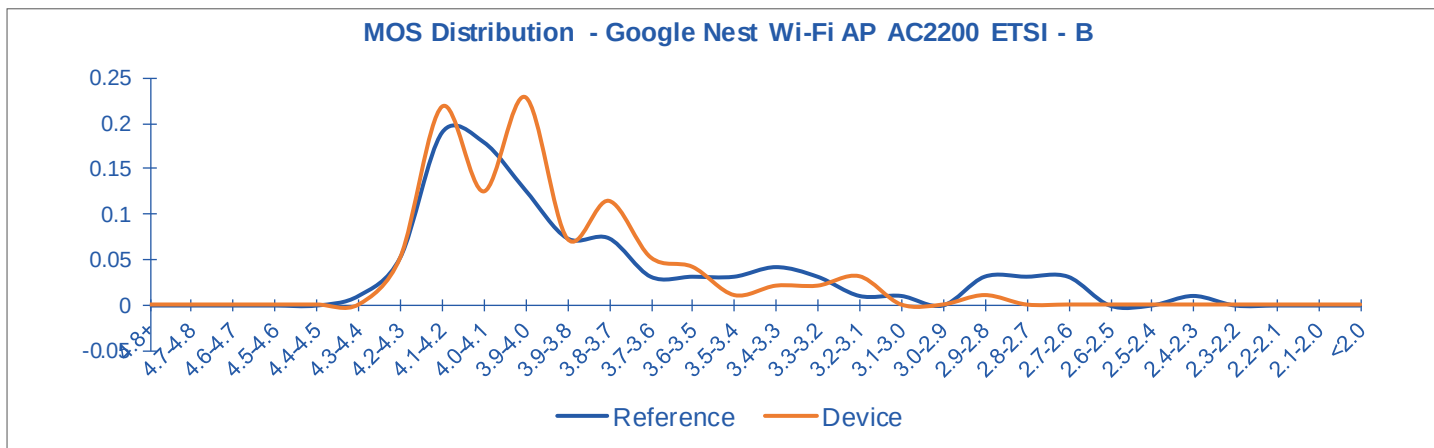
| WFC Call Performance - Google Nest Wi-Fi AP AC2200 |             |          |                     |                     |                        |                    |                  |               |      |
|----------------------------------------------------|-------------|----------|---------------------|---------------------|------------------------|--------------------|------------------|---------------|------|
| WFC                                                | Device Role | RSSI dBm | Connection Attempts | Mean Setup Time (s) | Successful Initiations | Failed Initiations | Successful Calls | Dropped Calls | MOS  |
| ETSI-B                                             | Reference   | -65      | 24                  | 6.63                | 100.0%                 | 0                  | 100.0%           | 0             | 3.87 |
| ETSI-B                                             | Device      | -65      | 24                  | 5.35                | 100.0%                 | 0                  | 100.0%           | 0             | 3.70 |
| NSD-A                                              | Reference   | -65      | 24                  | 6.57                | 100.0%                 | 0                  | 100.0%           | 0             | 3.63 |
| NSD-A                                              | Device      | -65      | 24                  | 5.33                | 100.0%                 | 0                  | 100.0%           | 0             | 3.75 |
| NSD-C                                              | Reference   | -65      | 24                  | 8.04                | 100.0%                 | 0                  | 100.0%           | 0             | 3.12 |
| NSD-C                                              | Device      | -65      | 24                  | 6.27                | 100.0%                 | 0                  | 100.0%           | 0             | 3.02 |

Note: All the values are from Mobile Origination (MO)



## WFC Call Performance

### Google Nest Wi-Fi AP AC2200

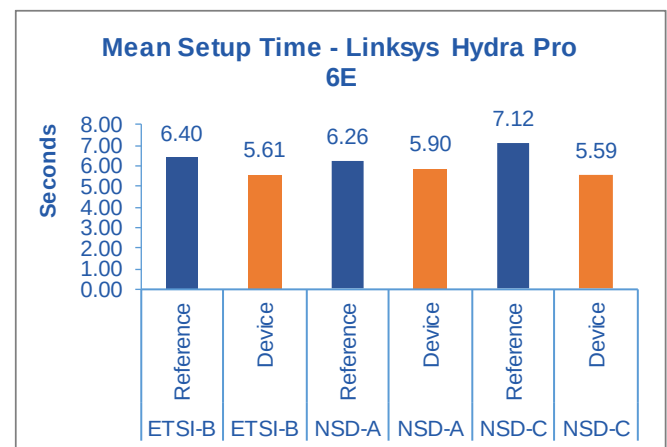
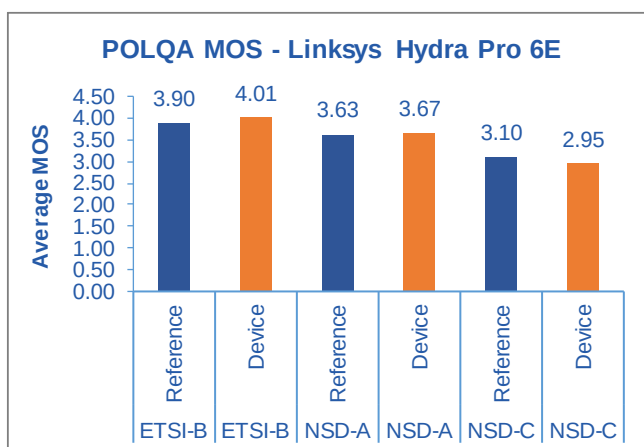
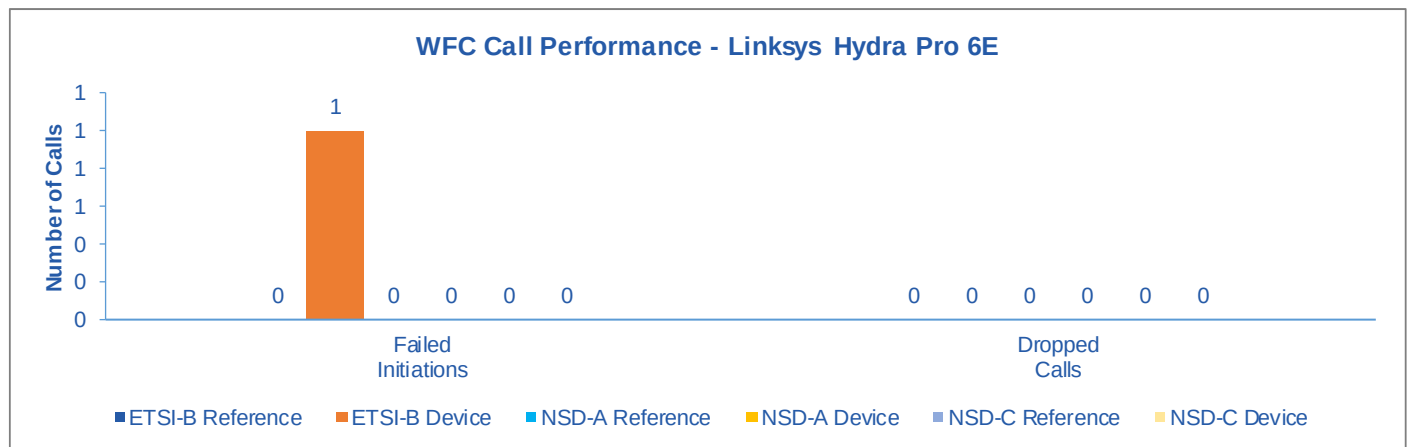


## WFC Call Performance

### Linksys Hydra Pro 6E

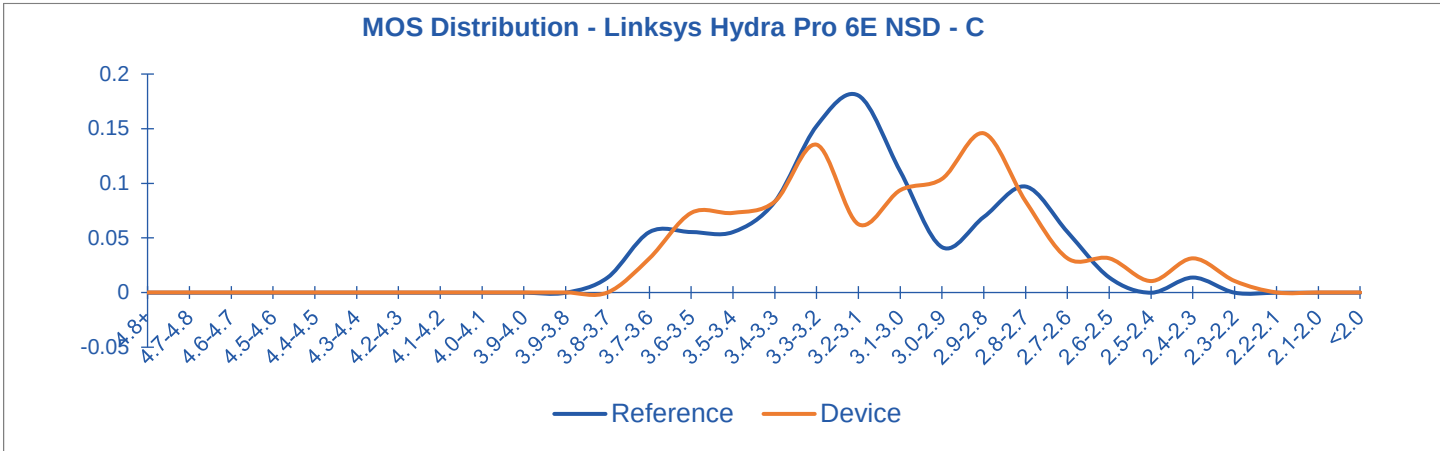
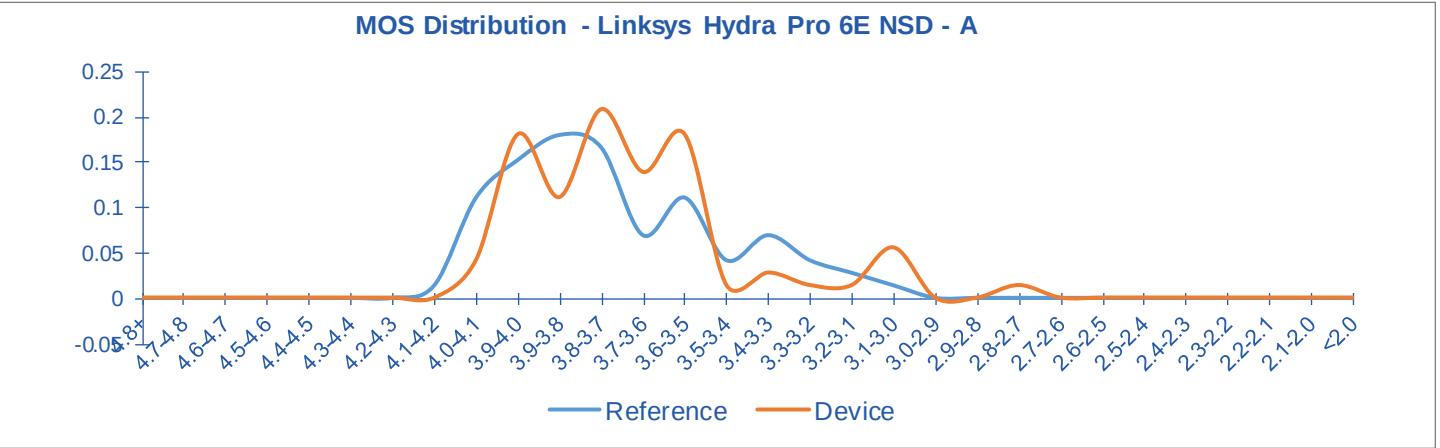
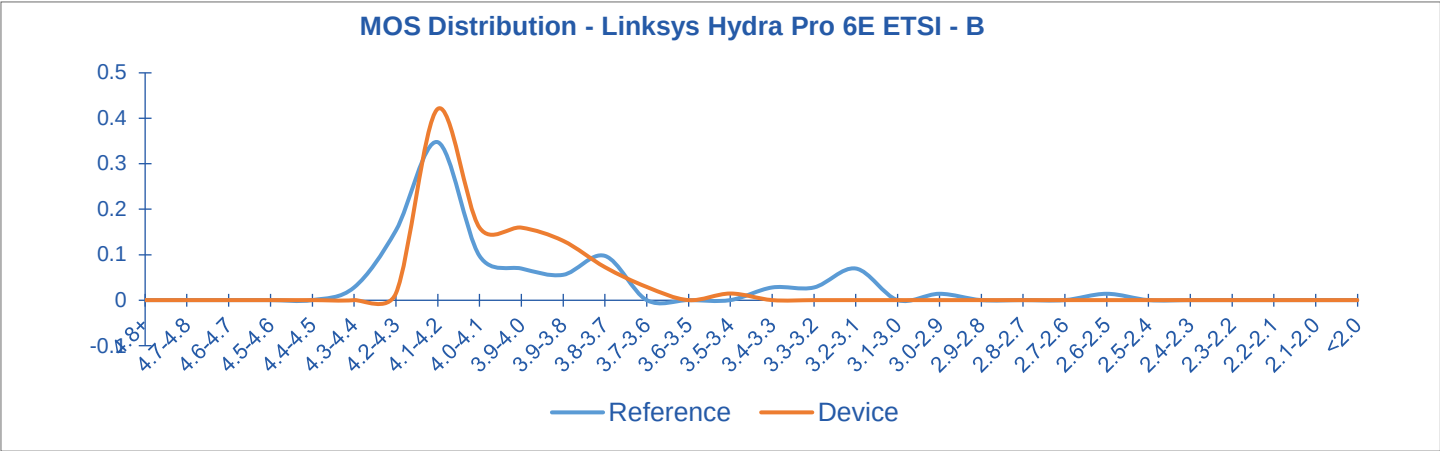
| WFC Call Performance - Linksys Hydra Pro 6E |             |          |                     |                     |                        |                    |                  |               |      |
|---------------------------------------------|-------------|----------|---------------------|---------------------|------------------------|--------------------|------------------|---------------|------|
| WFC                                         | Device Role | RSSI dBm | Connection Attempts | Mean Setup Time (s) | Successful Initiations | Failed Initiations | Successful Calls | Dropped Calls | MOS  |
| ETSI-B                                      | Reference   | -65      | 24                  | 6.40                | 100.0%                 | 0                  | 100.0%           | 0             | 3.90 |
| ETSI-B                                      | Device      | -65      | 24                  | 5.61                | 95.8%                  | 1                  | 100.0%           | 0             | 4.01 |
| NSD-A                                       | Reference   | -65      | 24                  | 6.26                | 100.0%                 | 0                  | 100.0%           | 0             | 3.63 |
| NSD-A                                       | Device      | -65      | 24                  | 5.90                | 100.0%                 | 0                  | 100.0%           | 0             | 3.67 |
| NSD-C                                       | Reference   | -65      | 24                  | 7.12                | 100.0%                 | 0                  | 100.0%           | 0             | 3.10 |
| NSD-C                                       | Device      | -65      | 24                  | 5.59                | 100.0%                 | 0                  | 100.0%           | 0             | 2.95 |

Note: All the values are from Mobile Origination (MO)



WFC Call Performance

Linksys Hydra Pro 6E



## Overall Handover Performance

## Handover - Overall

| Overall HandOver | Call Drops | Attempts | Max Attempts |
|------------------|------------|----------|--------------|
| Reference        | 0          | 27       | 36           |
| Device           | 1          | 22       | 36           |

There are 36 max attempts for Handover test cases and 18 are possible as minimum/best-case scenario.

| Multi Handover       |           | Call Drops | Max Attempts |
|----------------------|-----------|------------|--------------|
| 5G21-12W-A (Nokia)   | Reference | 0          | 5            |
| 5G21-12W-A (Nokia)   | Device    | 0          | 3            |
| ASUS RT-AC68U        | Reference | 0          | 4            |
| ASUS RT-AC68U        | Device    | 0          | 3            |
| Linksys Hydra Pro 6E | Reference | 0          | 4            |
| Linksys Hydra Pro 6E | Device    | 0          | 4            |

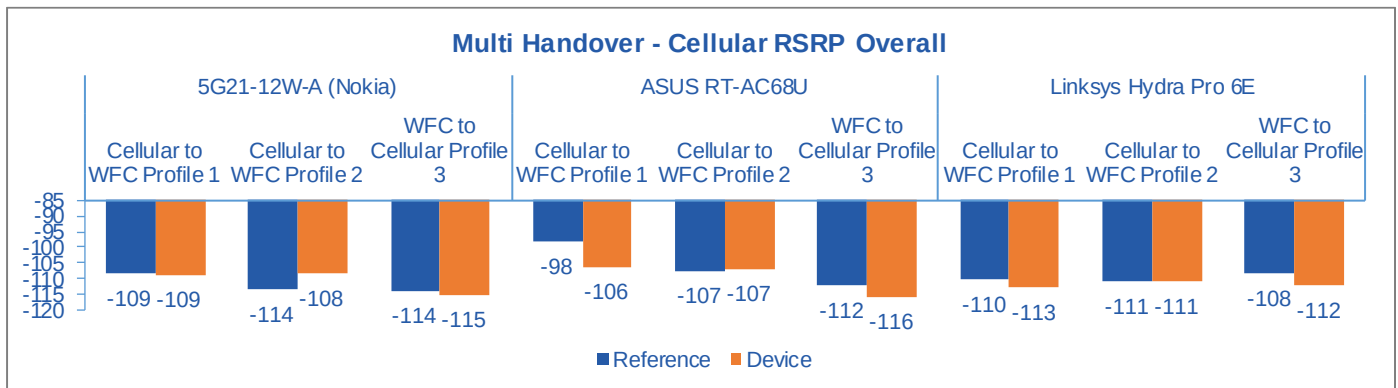
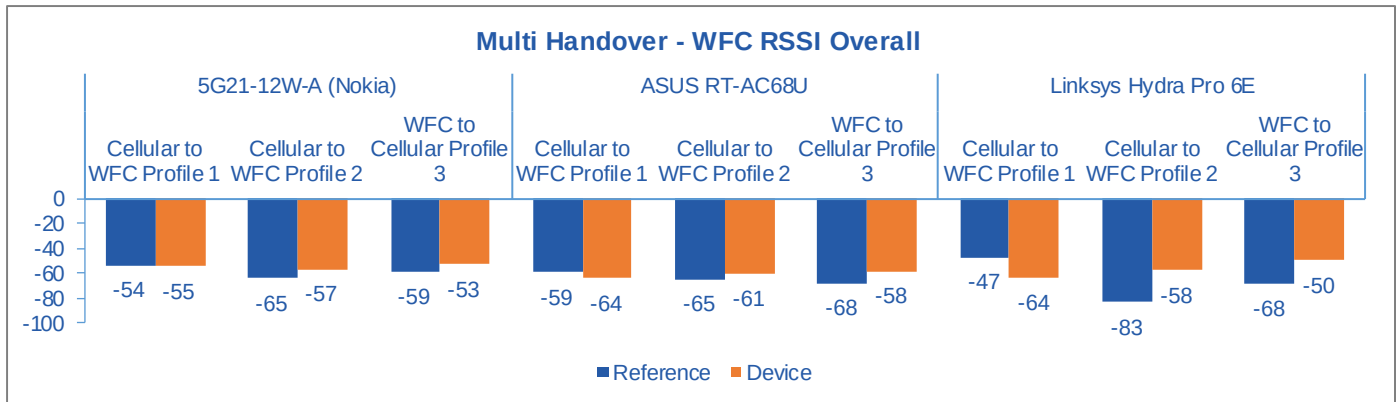
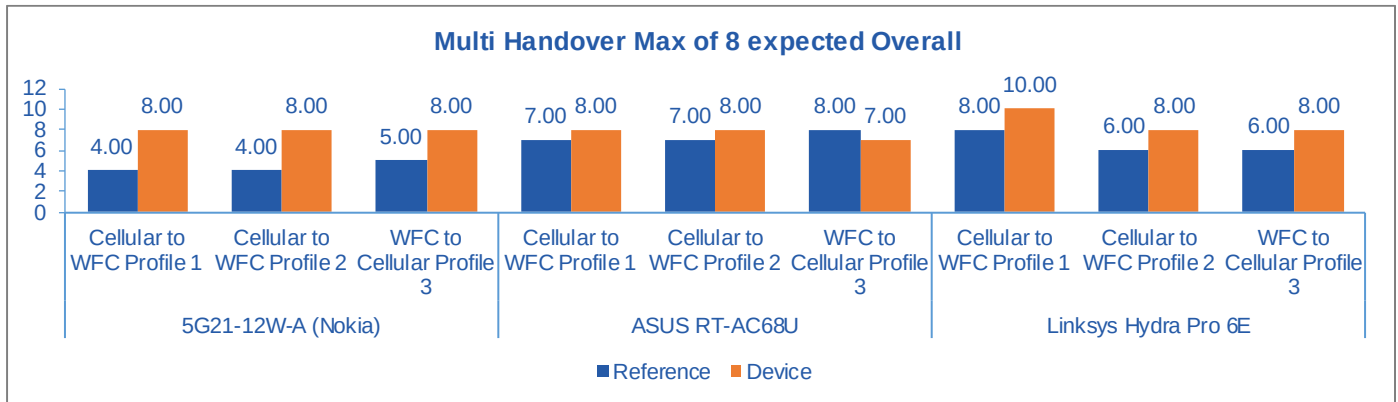
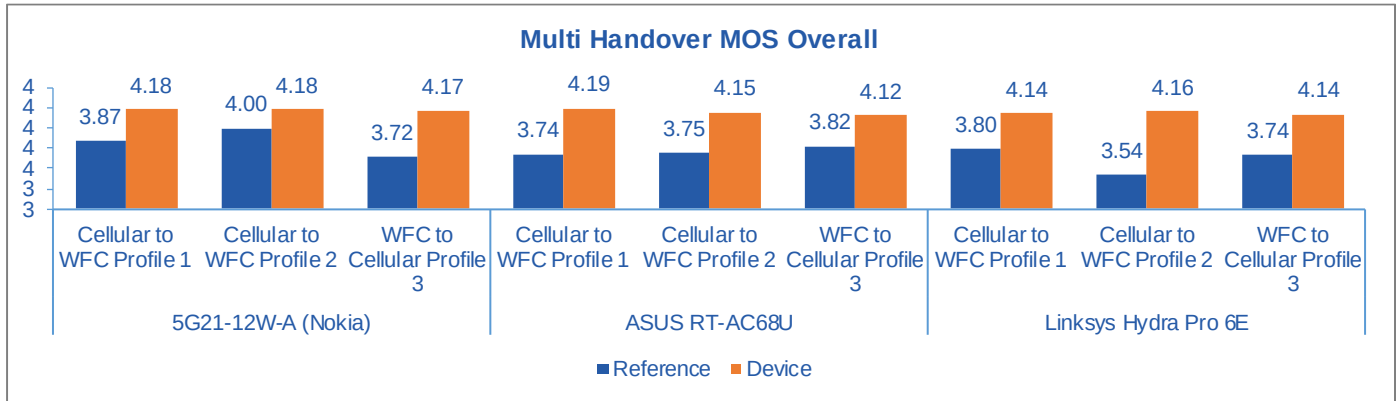
| WFC to Cellular             |           | Call Drops | Max Attempts |
|-----------------------------|-----------|------------|--------------|
| Google Nest Wi-Fi AP AC2200 | Reference | 0          | 2            |
| Google Nest Wi-Fi AP AC2200 | Device    | 0          | 2            |
| ASUS RT-AC68U               | Reference | 0          | 2            |
| ASUS RT-AC68U               | Device    | 0          | 2            |
| Linksys Hydra Pro 6E        | Reference | 0          | 2            |
| Linksys Hydra Pro 6E        | Device    | 1          | 2            |

| Walk In/Out of WFC Coverage |           | Call Drops | Max Attempts |
|-----------------------------|-----------|------------|--------------|
| 5G21-12W-A (Nokia)          | Reference | 0          | 4            |
| 5G21-12W-A (Nokia)          | Device    | 0          | 2            |
| ASUS RT-AC68U               | Reference | 0          | 2            |
| ASUS RT-AC68U               | Device    | 0          | 2            |
| Linksys Hydra Pro 6E        | Reference | 0          | 2            |
| Linksys Hydra Pro 6E        | Device    | 0          | 2            |

## Multi Handover



## Multi Handover - Overall



## Multi Handover

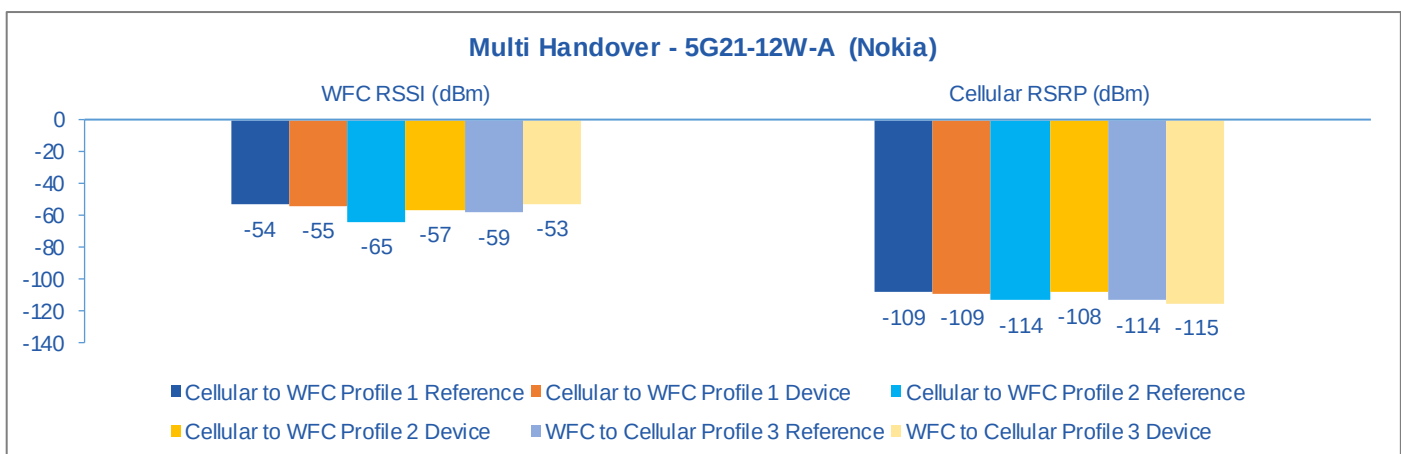
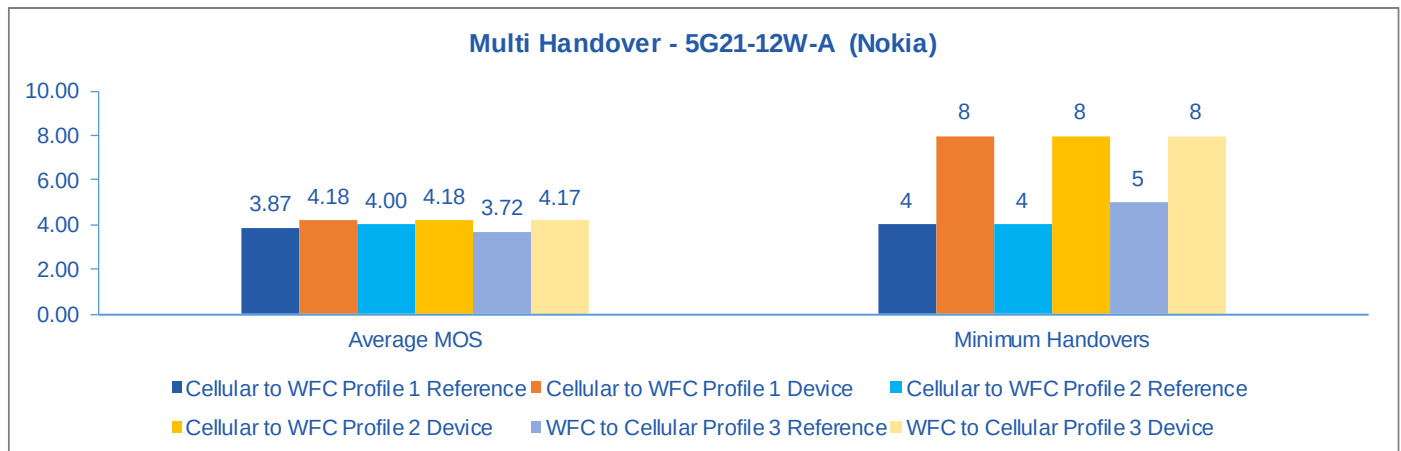
### 5G21-12W-A (Nokia)

| Multi Handover - 5G21-12W-A (Nokia) |             |             |                |                     |            |                   |                           |
|-------------------------------------|-------------|-------------|----------------|---------------------|------------|-------------------|---------------------------|
| Handover Test Case                  | Device Role | Average MOS | WFC RSSI (dBm) | Cellular RSRP (dBm) | Call Drops | Minimum Handovers | Attempts Max of 2 allowed |
| Cellular to WFC Profile 1           | Reference   | 3.87        | -54            | -109                | 0          | 4                 | 2                         |
| Cellular to WFC Profile 1           | Device      | 4.18        | -55            | -109                | 0          | 8                 | 1                         |
| Cellular to WFC Profile 2           | Reference   | 4.00        | -65            | -114                | 0          | 4                 | 2                         |
| Cellular to WFC Profile 2           | Device      | 4.18        | -57            | -108                | 0          | 8                 | 1                         |
| WFC to Cellular Profile 3           | Reference   | 3.72        | -59            | -114                | 0          | 5                 | 1                         |
| WFC to Cellular Profile 3           | Device      | 4.17        | -53            | -115                | 0          | 8                 | 1                         |

Note: No values indicated device failed to execute testcase.

No data represents call drop or device held onto call to signal call was initiated from.

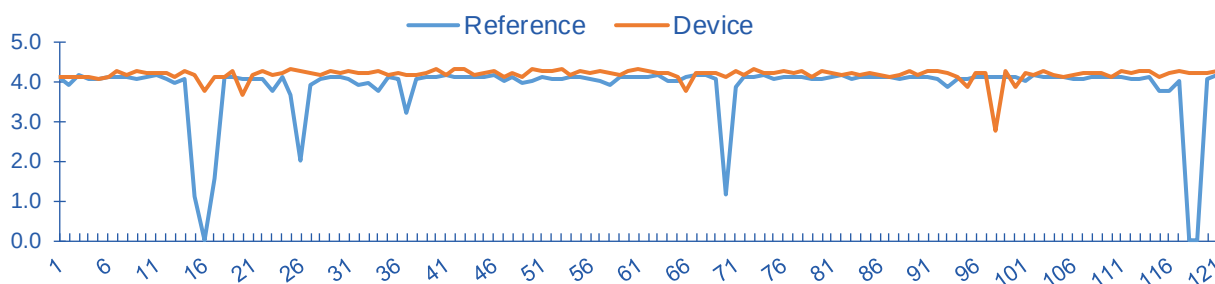
MOS before handover represents last 10 MOS samples before call dropped or test ended without handover



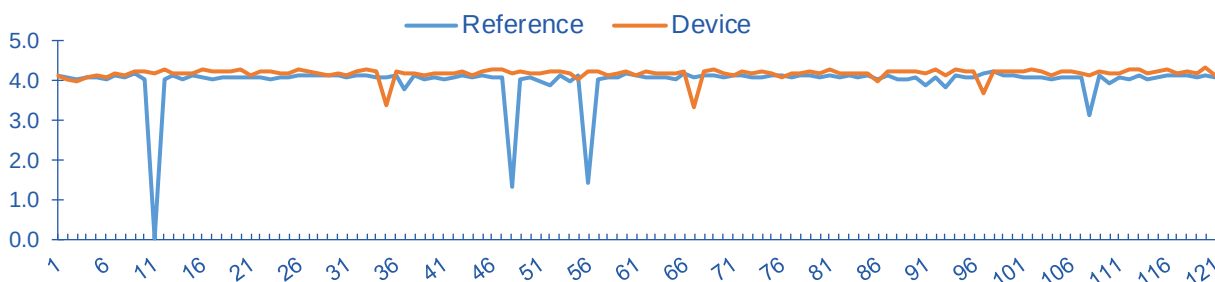
## Multi Handover

### 5G21-12W-A (Nokia)

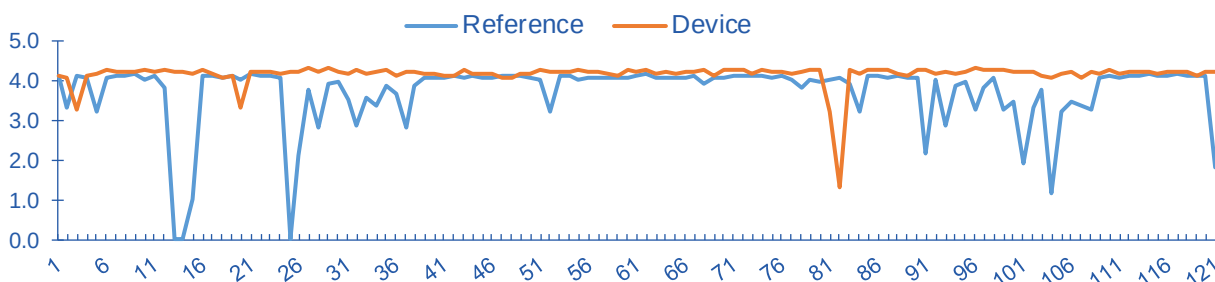
Multi Handover 5G21-12W-A (Nokia) - Profile 1 MOS Scores



Multi Handover 5G21-12W-A (Nokia) - Profile 2 MOS Scores



Multi Handover 5G21-12W-A (Nokia) - Profile 3 MOS Scores



## Multi Handover

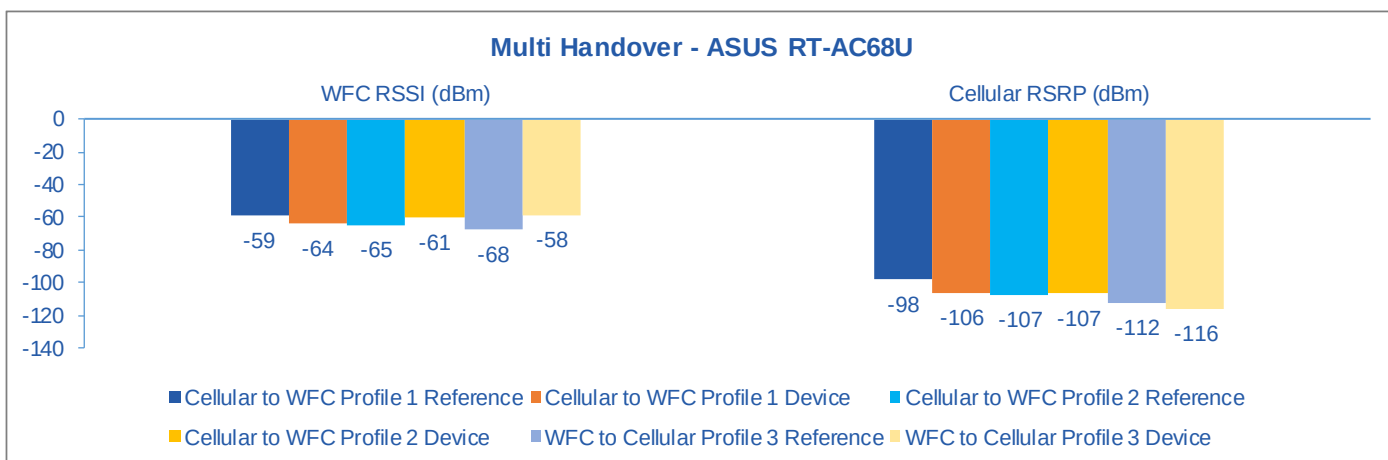
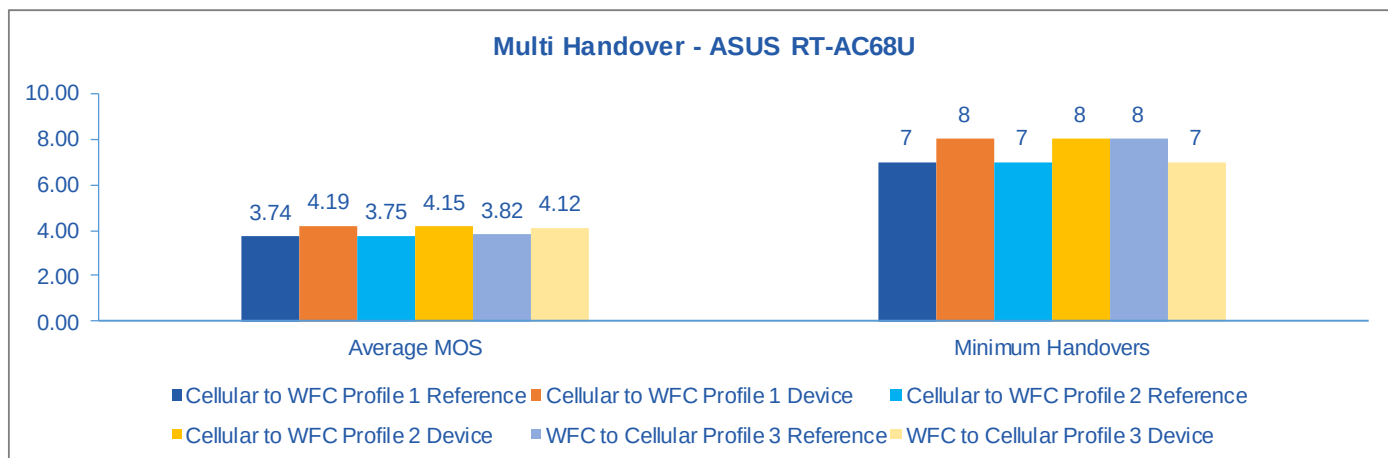
### ASUS RT-AC68U

| Multi Handover - ASUS RT-AC68U |             |             |                |                     |            |                   |                           |
|--------------------------------|-------------|-------------|----------------|---------------------|------------|-------------------|---------------------------|
| Handover Test Case             | Device Role | Average MOS | WFC RSSI (dBm) | Cellular RSRP (dBm) | Call Drops | Minimum Handovers | Attempts Max of 2 allowed |
| Cellular to WFC Profile 1      | Reference   | 3.74        | -59            | -98                 | 0          | 7                 | 1                         |
| Cellular to WFC Profile 1      | Device      | 4.19        | -64            | -106                | 0          | 8                 | 1                         |
| Cellular to WFC Profile 2      | Reference   | 3.75        | -65            | -107                | 0          | 7                 | 1                         |
| Cellular to WFC Profile 2      | Device      | 4.15        | -61            | -107                | 0          | 8                 | 1                         |
| WFC to Cellular Profile 3      | Reference   | 3.82        | -68            | -112                | 0          | 8                 | 2                         |
| WFC to Cellular Profile 3      | Device      | 4.12        | -58            | -116                | 0          | 7                 | 1                         |

Note: No values indicated device failed to execute testcase.

No data represents call drop or device held onto call to signal call was initiated from.

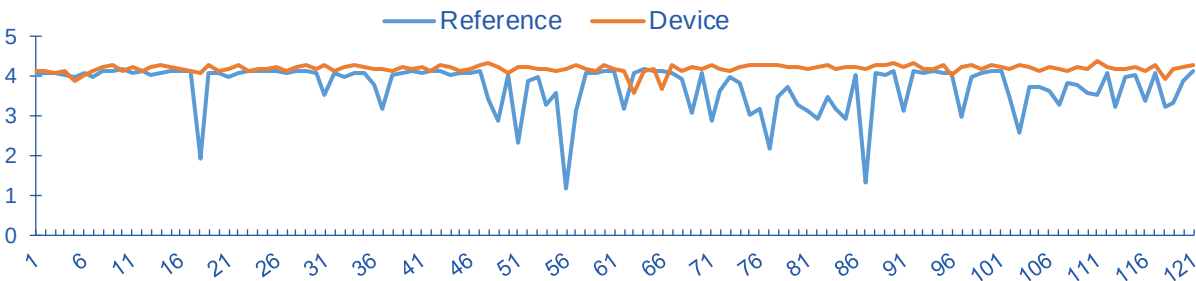
MOS before handover represents last 10 MOS samples before call dropped or test ended without handover



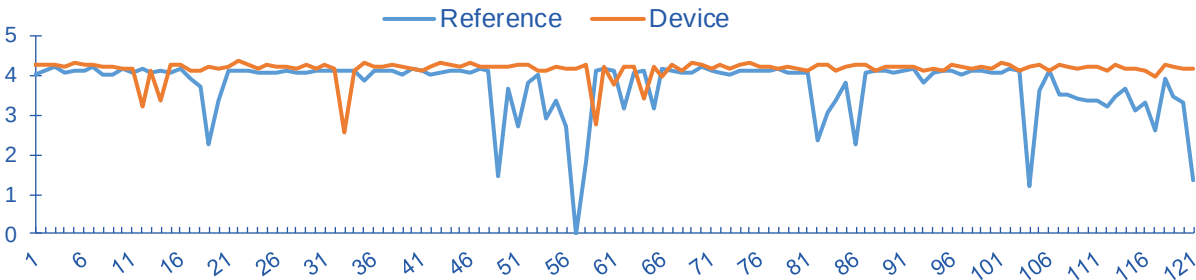
**Multi Handover**

ASUS RT-AC68U

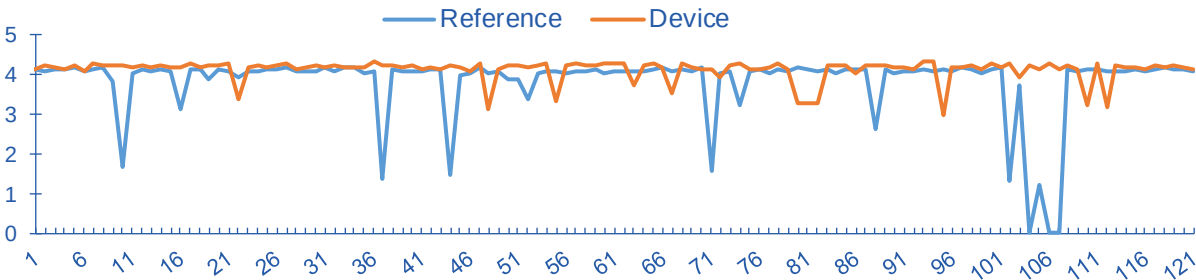
Multi Handover ASUS RT-AC68U - Profile 1 MOS Scores



Multi Handover ASUS RT-AC68U - Profile 2 MOS Scores



Multi Handover ASUS RT-AC68U - Profile 3 MOS Scores



## Multi Handover

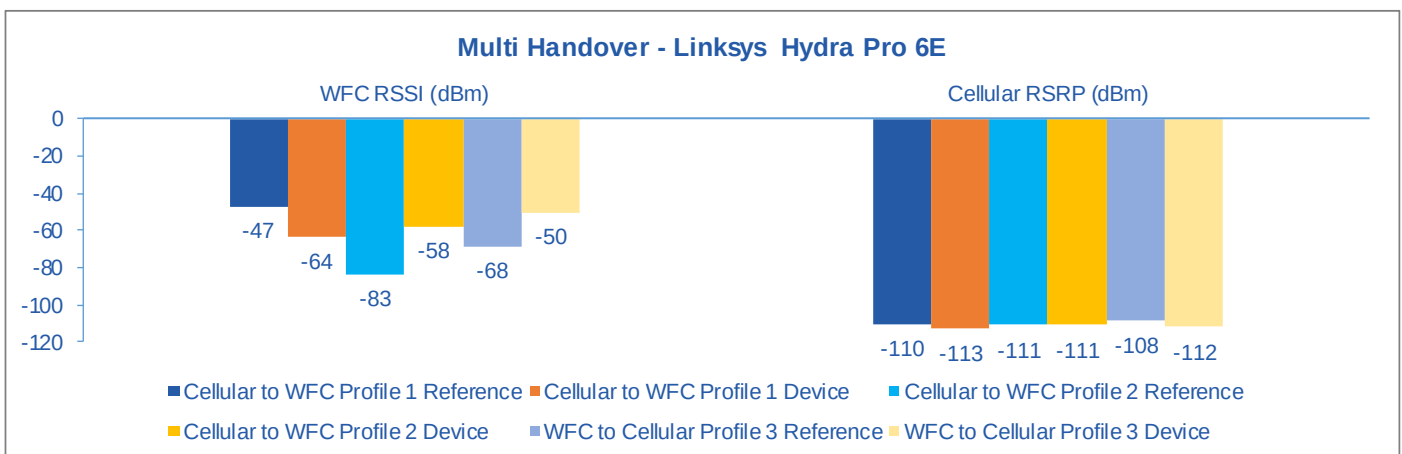
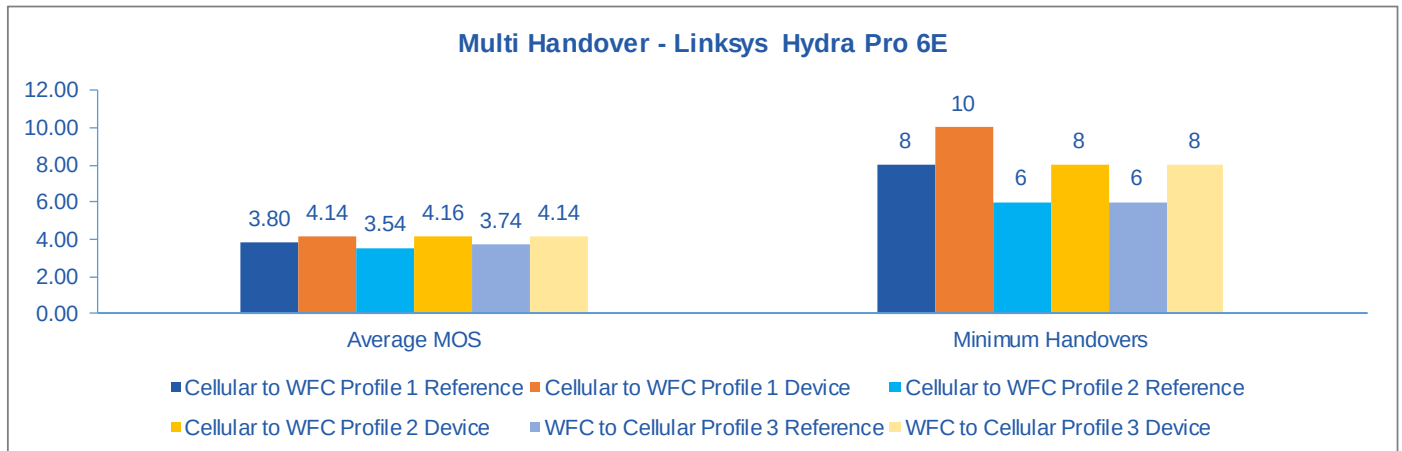
### Linksys Hydra Pro 6E

| Multi Handover - Linksys Hydra Pro 6E |             |             |                |                     |            |                   |                           |
|---------------------------------------|-------------|-------------|----------------|---------------------|------------|-------------------|---------------------------|
| Handover Test Case                    | Device Role | Average MOS | WFC RSSI (dBm) | Cellular RSRP (dBm) | Call Drops | Minimum Handovers | Attempts Max of 2 allowed |
| Cellular to WFC Profile 1             | Reference   | 3.80        | -47            | -110                | 0          | 8                 | 2                         |
| Cellular to WFC Profile 1             | Device      | 4.14        | -64            | -113                | 0          | 10                | 1                         |
| Cellular to WFC Profile 2             | Reference   | 3.54        | -83            | -111                | 0          | 6                 | 1                         |
| Cellular to WFC Profile 2             | Device      | 4.16        | -58            | -111                | 0          | 8                 | 1                         |
| WFC to Cellular Profile 3             | Reference   | 3.74        | -68            | -108                | 0          | 6                 | 1                         |
| WFC to Cellular Profile 3             | Device      | 4.14        | -50            | -112                | 0          | 8                 | 2                         |

Note: No values indicated device failed to execute testcase.

No data represents call drop or device held onto call to signal call was initiated from.

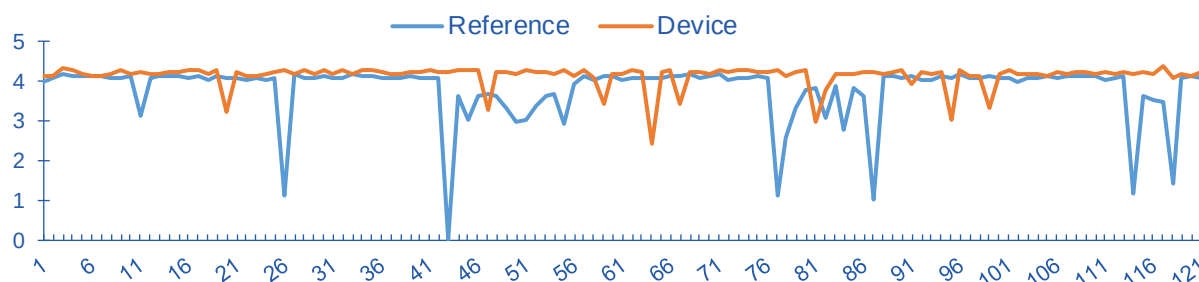
MOS before handover represents last 10 MOS samples before call dropped or test ended without handover



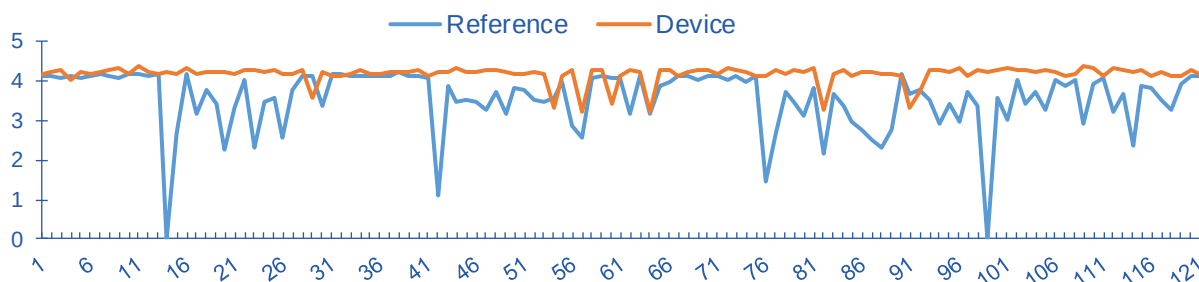
## Multi Handover

### Linksys Hydra Pro 6E

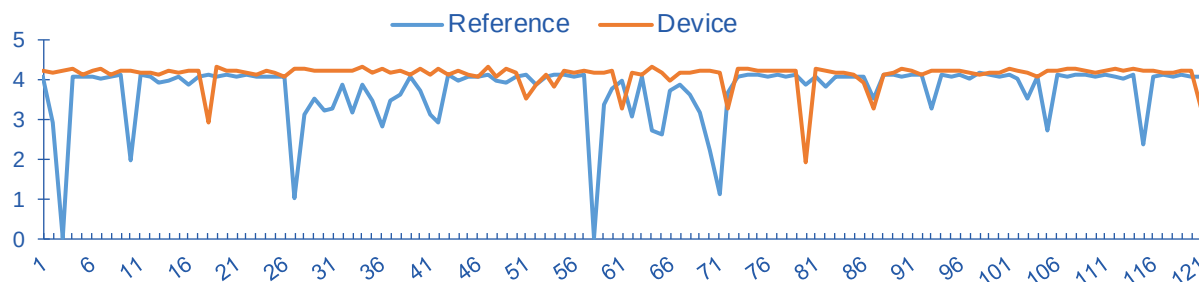
Multi Handover Linksys Hydra Pro 6E - Profile 1 MOS Scores



Multi Handover Linksys Hydra Pro 6E - Profile 2 MOS Scores



Multi Handover Linksys Hydra Pro 6E - Profile 3 MOS Scores



## IP Impairments



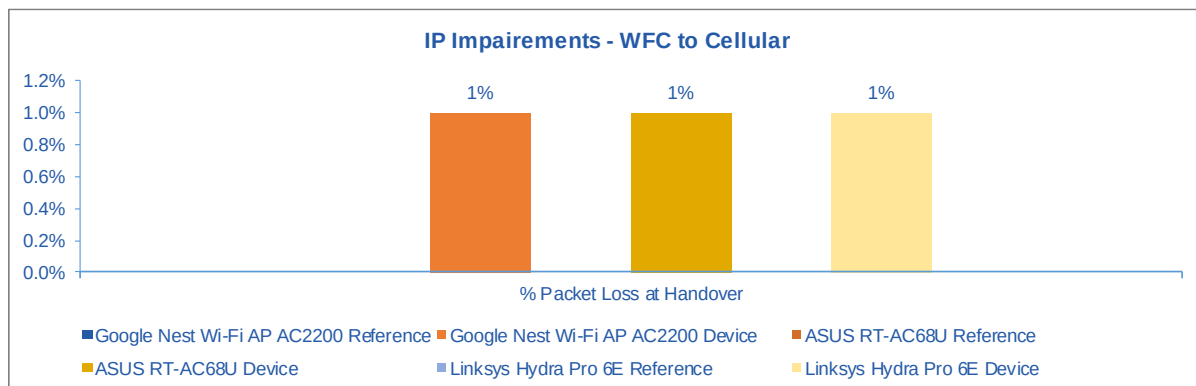
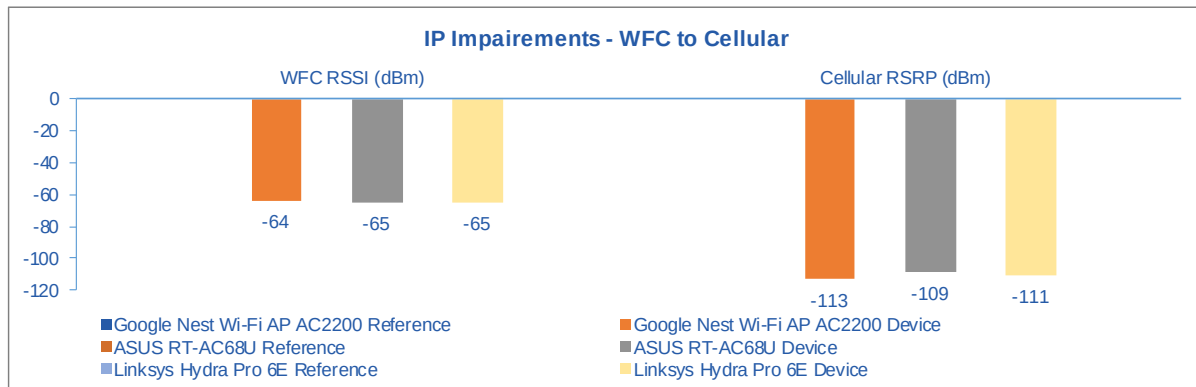
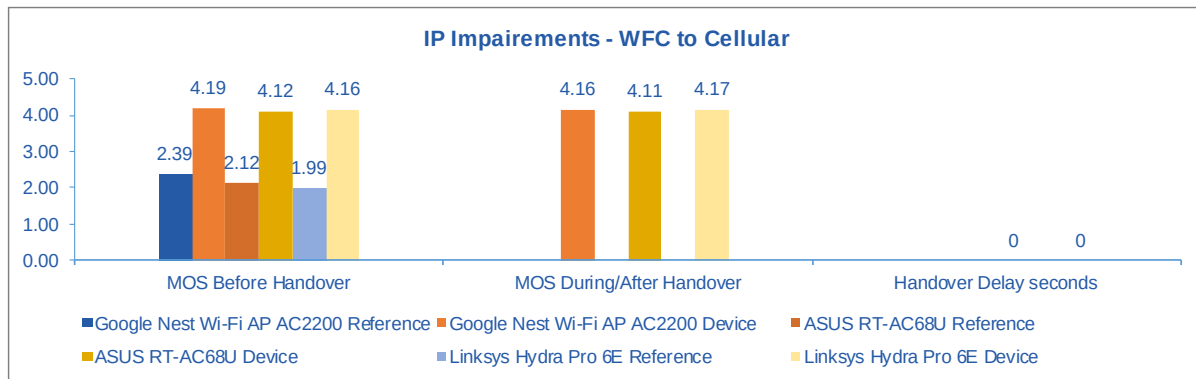
## IP Impairments (Profile 4)

| WFC to Cellular                       | MOS Before Handover | MOS During/After Handover | % Packet Loss at Handover | Jitter at Handover | Handover Delay seconds | WFC RSSI (dBm) | Cellular RSRP (dBm) | Call Drops | Attempts Max of 2 allowed |
|---------------------------------------|---------------------|---------------------------|---------------------------|--------------------|------------------------|----------------|---------------------|------------|---------------------------|
| Google Nest Wi-Fi AP AC2200 Reference | 2.39                |                           |                           |                    |                        |                |                     | 0          | 2                         |
| Google Nest Wi-Fi AP AC2200 Device    | 4.19                | 4.16                      | 1%                        | 2                  |                        | -64            | -113                | 0          | 2                         |
| ASUS RT-AC68U Reference               | 2.12                |                           |                           |                    |                        |                |                     | 0          | 2                         |
| ASUS RT-AC68U Device                  | 4.12                | 4.11                      | 1%                        | 2                  | 0                      | -65            | -109                | 0          | 2                         |
| Linksys Hydra Pro 6E Reference        | 1.99                |                           |                           |                    |                        |                |                     | 0          | 2                         |
| Linksys Hydra Pro 6E Device           | 4.16                | 4.17                      | 1%                        | 2                  | 0                      | -65            | -111                | 1          | 2                         |

Note: No values indicated device failed to execute testcase.

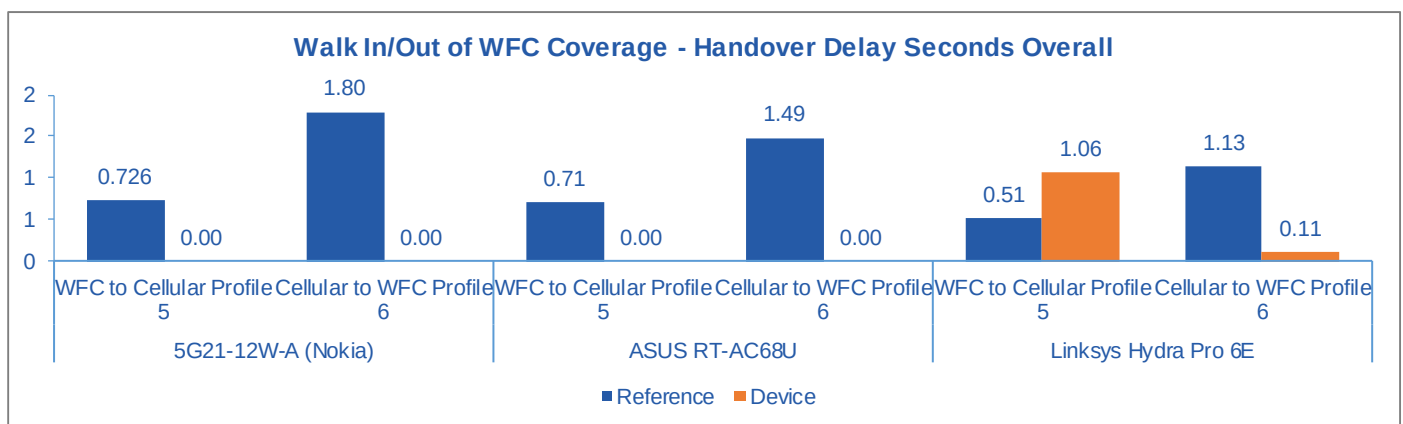
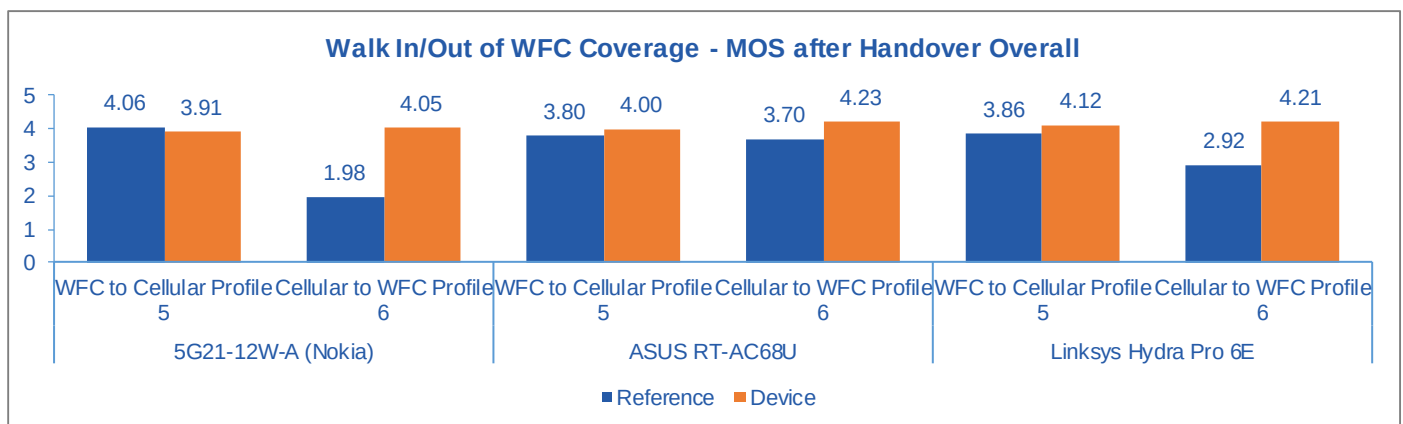
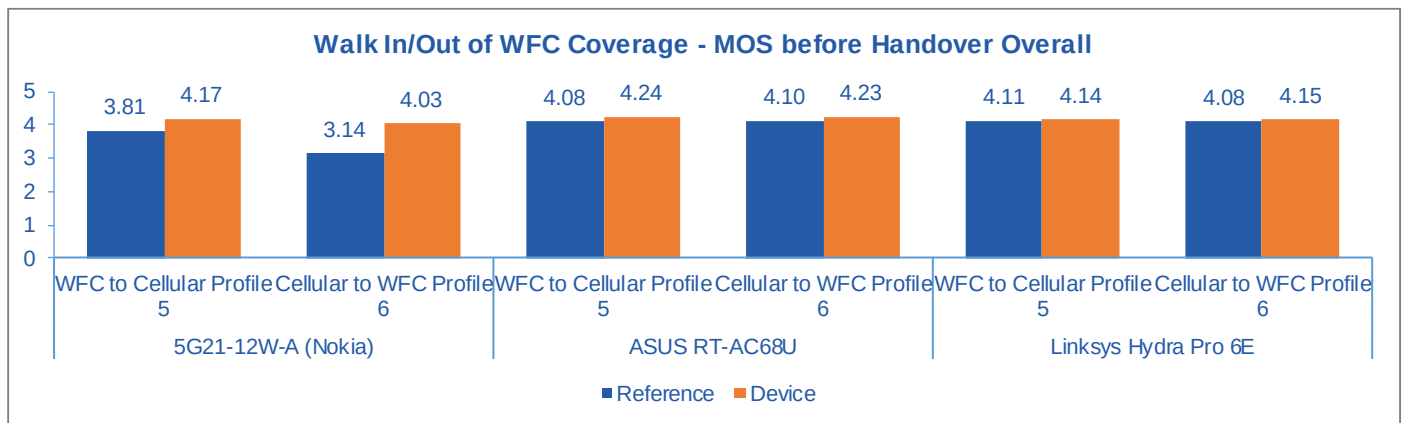
No data represents call drop or device held onto call to signal call was initiated from.

MOS before handover represents last 10 MOS samples before call dropped or test ended without handover

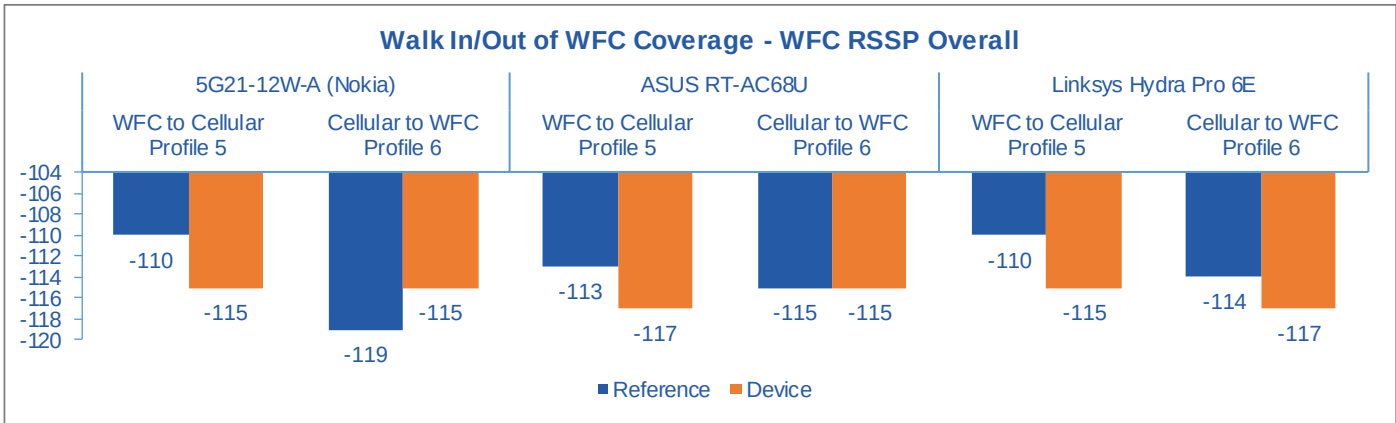
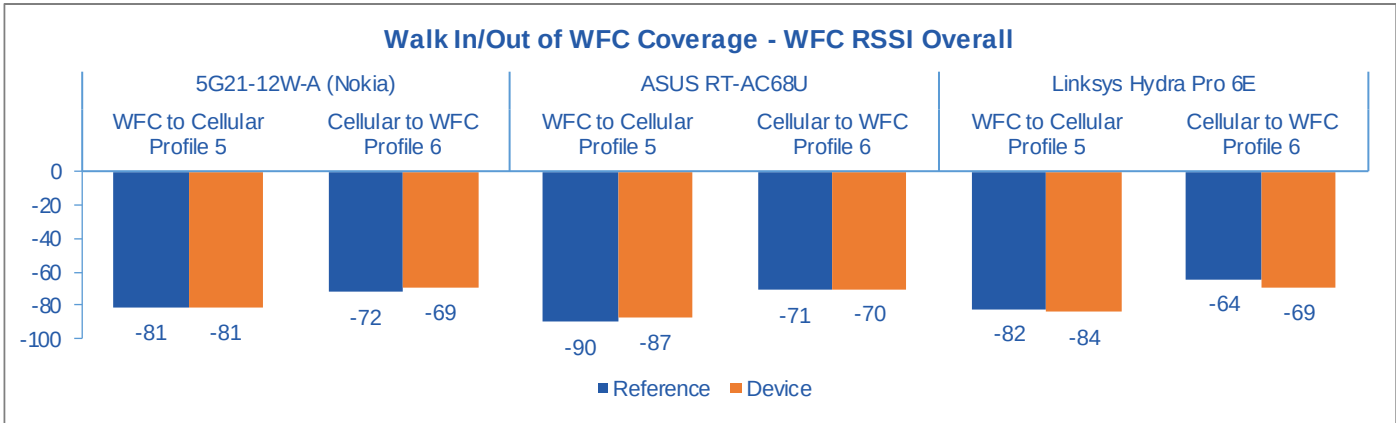


## Walk In/Out of WFC Coverage

## Walk In/Out of WFC Coverage - Overall



Walk In/Out of WFC Coverage - Overall



## Walk In/Out of WFC Coverage

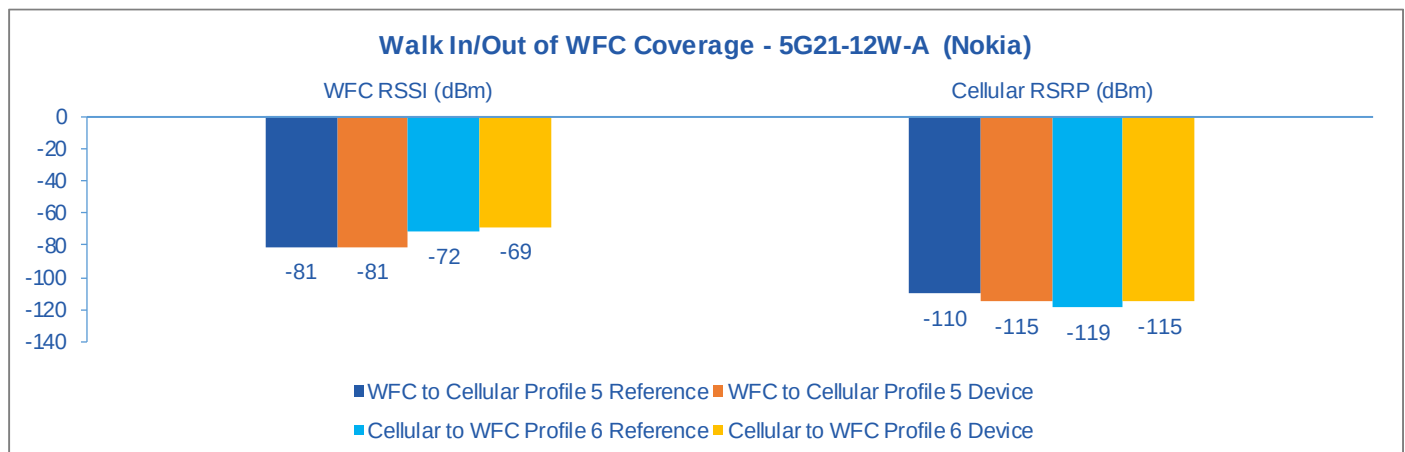
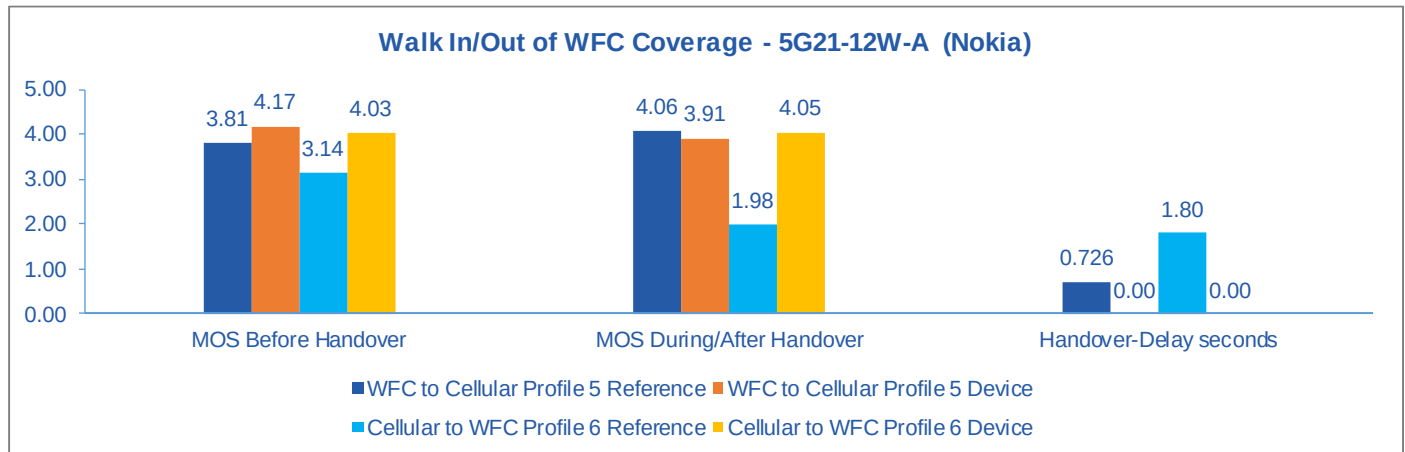
### 5G21-12W-A (Nokia)

| Walk In/Out of WFC Coverage - 5G21-12W-A (Nokia) |             |                     |                           |                        |                |                     |            |                           |
|--------------------------------------------------|-------------|---------------------|---------------------------|------------------------|----------------|---------------------|------------|---------------------------|
| Handover Test Case                               | Device Role | MOS Before Handover | MOS During/After Handover | Handover-Delay seconds | WFC RSSI (dBm) | Cellular RSRP (dBm) | Call Drops | Attempts Max of 2 allowed |
| WFC to Cellular Profile 5                        | Reference   | 3.81                | 4.06                      | 0.726                  | -81            | -110                | 0          | 2                         |
| WFC to Cellular Profile 5                        | Device      | 4.17                | 3.91                      | 0.00                   | -81            | -115                | 0          | 1                         |
| Cellular to WFC Profile 6                        | Reference   | 3.14                | 1.98                      | 1.80                   | -72            | -119                | 0          | 2                         |
| Cellular to WFC Profile 6                        | Device      | 4.03                | 4.05                      | 0.00                   | -69            | -115                | 0          | 1                         |

Note: No values indicated device failed to execute testcase.

No data represents call drop or device held onto call to signal call was initiated from.

MOS before handover represents last 10 MOS samples before call dropped or test ended without handover



## Walk In/Out of WFC Coverage

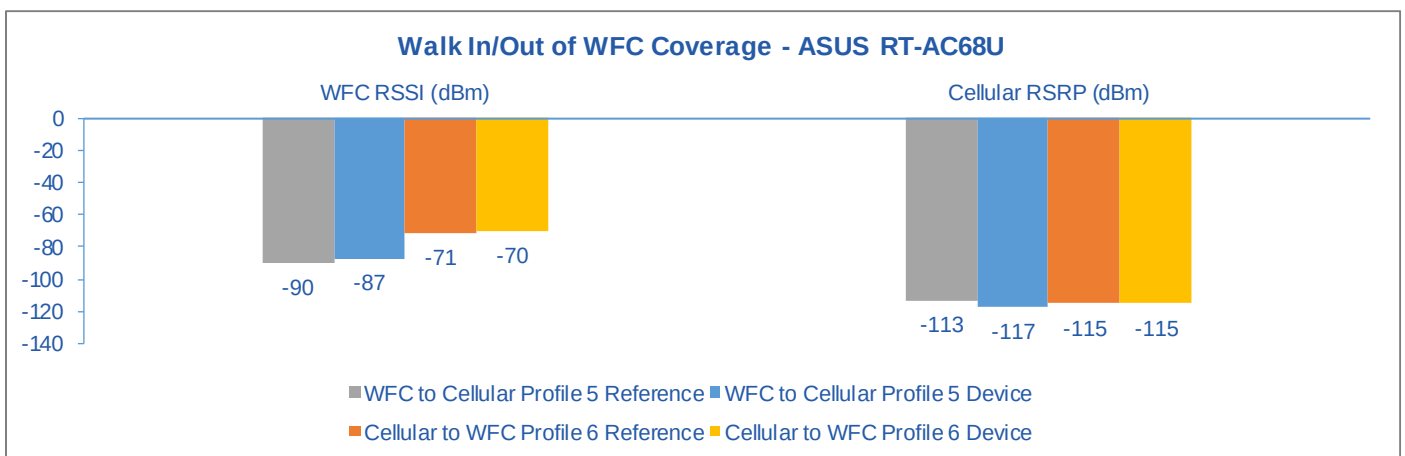
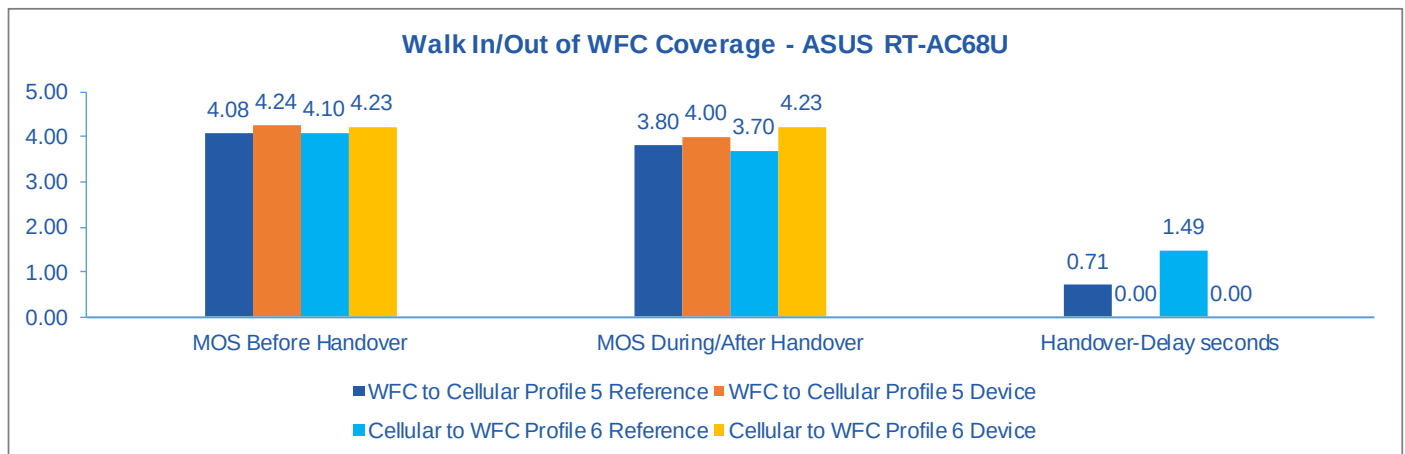
### ASUS RT-AC68U

| Walk In/Out of WFC Coverage - ASUS RT-AC68U |             |                     |                           |                        |                |                     |            |                           |
|---------------------------------------------|-------------|---------------------|---------------------------|------------------------|----------------|---------------------|------------|---------------------------|
| Handover Test Case                          | Device Role | MOS Before Handover | MOS During/After Handover | Handover-Delay seconds | WFC RSSI (dBm) | Cellular RSRP (dBm) | Call Drops | Attempts Max of 2 allowed |
| WFC to Cellular Profile 5                   | Reference   | 4.08                | 3.80                      | 0.71                   | -90            | -113                | 0          | 1                         |
| WFC to Cellular Profile 5                   | Device      | 4.24                | 4.00                      | 0.00                   | -87            | -117                | 0          | 1                         |
| Cellular to WFC Profile 6                   | Reference   | 4.10                | 3.70                      | 1.49                   | -71            | -115                | 0          | 1                         |
| Cellular to WFC Profile 6                   | Device      | 4.23                | 4.23                      | 0.00                   | -70            | -115                | 0          | 1                         |

Note: No values indicated device failed to execute testcase.

No data represents call drop or device held onto call to signal call was initiated from.

MOS before handover represents last 10 MOS samples before call dropped or test ended without handover



## Walk In/Out of WFC Coverage

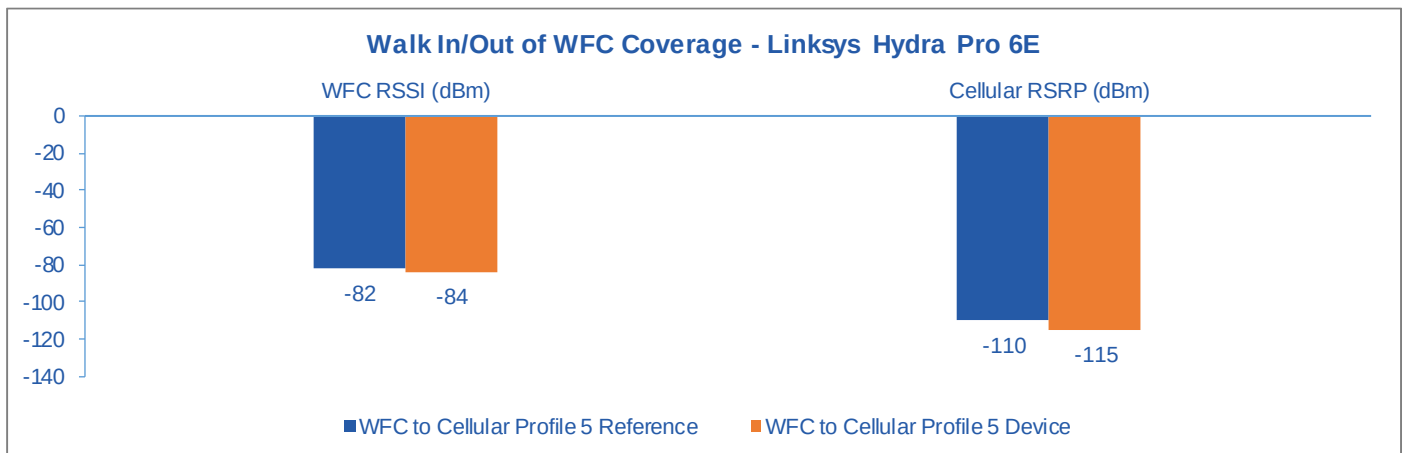
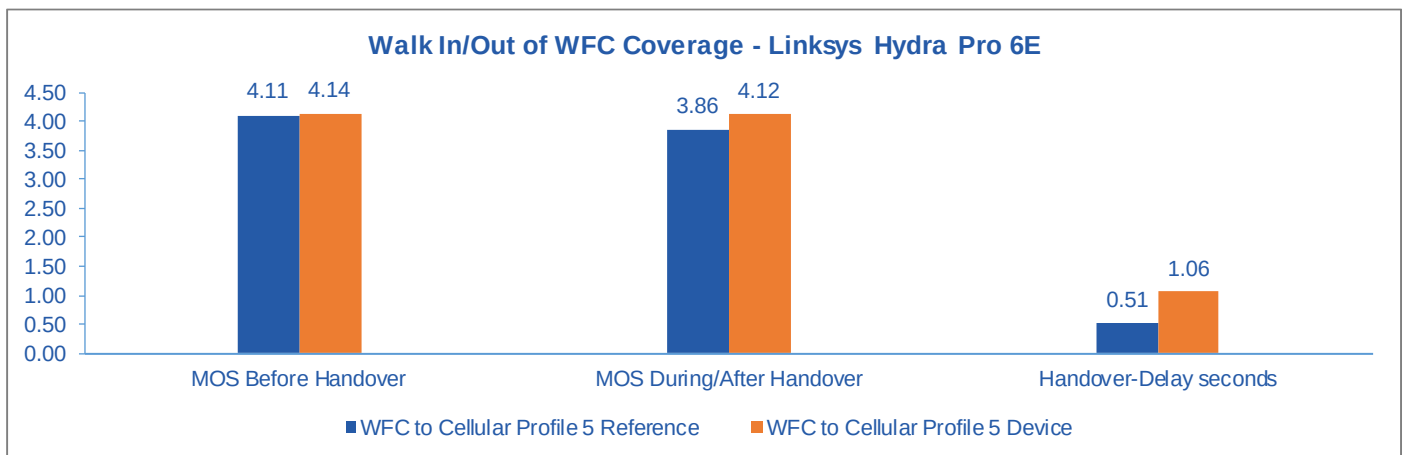
### Linksys Hydra Pro 6E

| Walk In/Out of WFC Coverage - Linksys Hydra Pro 6E |             |                     |                           |                        |                |                     |            |                           |
|----------------------------------------------------|-------------|---------------------|---------------------------|------------------------|----------------|---------------------|------------|---------------------------|
| Handover Test Case                                 | Device Role | MOS Before Handover | MOS During/After Handover | Handover-Delay seconds | WFC RSSI (dBm) | Cellular RSRP (dBm) | Call Drops | Attempts Max of 2 allowed |
| WFC to Cellular Profile 5                          | Reference   | 4.11                | 3.86                      | 0.51                   | -82            | -110                | 0          | 1                         |
| WFC to Cellular Profile 5                          | Device      | 4.14                | 4.12                      | 1.06                   | -84            | -115                | 0          | 1                         |
| Cellular to WFC Profile 6                          | Reference   | 4.08                | 2.92                      | 1.13                   | -64            | -114                | 0          | 1                         |
| Cellular to WFC Profile 6                          | Device      | 4.15                | 4.21                      | 0.11                   | -69            | -117                | 0          | 1                         |

Note: No values indicated device failed to execute testcase.

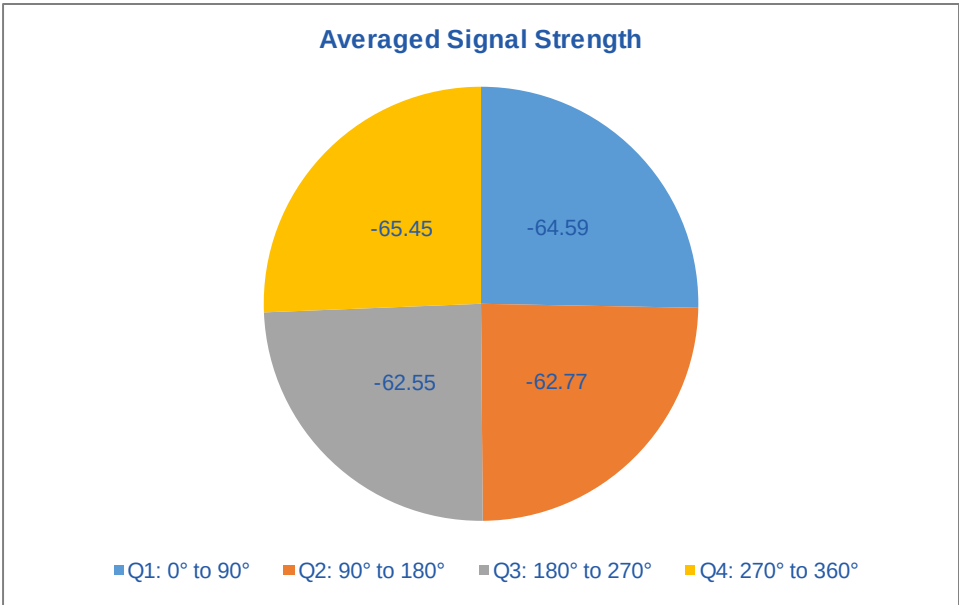
No data represents call drop or device held onto call to signal call was initiated from.

MOS before handover represents last 10 MOS samples before call dropped or test ended without handover



# WiFi Coverage Statistics

| Averaged Signal Strength |                 |                  |                  |
|--------------------------|-----------------|------------------|------------------|
| Q1: 0° to 90°            | Q2: 90° to 180° | Q3: 180° to 270° | Q4: 270° to 360° |
| -64.59                   | -62.77          | -62.55           | -65.45           |



Note : Turntable Quadrant at 0.23 RPM, Sampled every 3 Seconds



# Appendix

## Impaired WFC Test (WiFi Only)

- The purpose of this testing is to document devices WFC AFR/DCR during IP impairments and measure Audio Quality and performance.
- Each AP shall have IP impairments introducing packet loss, jitter and gaussian distribution introduced per ETSI TR 126 standard profiles and two non-standard profiles defined by T-Mobile.

- ETSI - Std Impairments

|          |                       |                 |                                   |
|----------|-----------------------|-----------------|-----------------------------------|
| <b>B</b> | Gaussian distribution | Packet Loss: 3% | Downlink Jitter: $\sigma = 40$ ms |
|----------|-----------------------|-----------------|-----------------------------------|

- T-Mobile Defined - Non Std Impairments

|          |                       |                  |                                   |
|----------|-----------------------|------------------|-----------------------------------|
| <b>A</b> | Gaussian distribution | Packet Loss: 8%  | Downlink Jitter: $\sigma = 10$ ms |
| <b>C</b> | Gaussian distribution | Packet Loss: 15% | Downlink Jitter: $\sigma = 60$ ms |

## Passing Requirements

| Call Performance ETSI-A, NSD-B, NSD-C |                                                             |
|---------------------------------------|-------------------------------------------------------------|
| Mean Setup Time                       | if(Avg of DUT) < (Avg of REF)                               |
|                                       | if(Avg of DUT) < (Avg of REF*1.1)                           |
|                                       | if(Avg of DUT) between (Avg of REF)*1.25 & 1.1              |
|                                       | if(Avg of DUT) > (Avg of REF*1.25)                          |
| Success/Drops                         | If p-value > 0.95 for DUT vs REF                            |
|                                       | If p-value between 0.05 and 0.95 for DUT vs REF             |
|                                       | RED condition is satisfied but % failures/Drops of DUT < 1% |
|                                       | If p-value for Fisher Exact < 0.05 for DUT vs REF           |
| MOS                                   | If DUT > REF                                                |
|                                       | If DUT > REF-0.1                                            |
|                                       | If Red or Green Not Satisfied                               |
|                                       | If DUT<REF-0.25                                             |

| RF Multi Handover |                               |
|-------------------|-------------------------------|
| MOS               | If DUT > REF                  |
|                   | If DUT > REF-0.1              |
|                   | If Red or Green Not Satisfied |
|                   | If DUT<REF-0.25               |
| WFC RSSI          | If WFC RSSI >= -73            |
|                   | If WFC RSSI < -73             |
| Cellular RSSP     | If Cellular RSRP >= -119      |
|                   | If Cellular RSRP < -119       |
| Minimum Handovers | If No of Handovers >=5        |
|                   | If No of Handovers < 5        |

| RF Handover WFC to Cellular |                               |
|-----------------------------|-------------------------------|
| MOS                         | If DUT > REF                  |
|                             | If DUT > REF-0.1              |
|                             | If Red or Green Not Satisfied |
|                             | If DUT<REF-0.25               |
| Handover Impact             | If Avg of Seconds <= 2        |
|                             | If Avg of Seconds > 2         |
| RSSI                        | If RSSI <= -77                |
|                             | If RSSI > -77                 |

## Passing Requirements

| RF Handover Cellular to WFC |                               |
|-----------------------------|-------------------------------|
| MOS                         | If DUT > REF                  |
|                             | If DUT > REF-0.1              |
|                             | If Red or Green Not Satisfied |
|                             | If DUT < REF-0.25             |
| Handover Impact             | If Avg of Seconds <= 2        |
|                             | If Avg of Seconds > 2         |
| RSSI                        | If RSSI >= -73                |
|                             | If RSSI < -73                 |

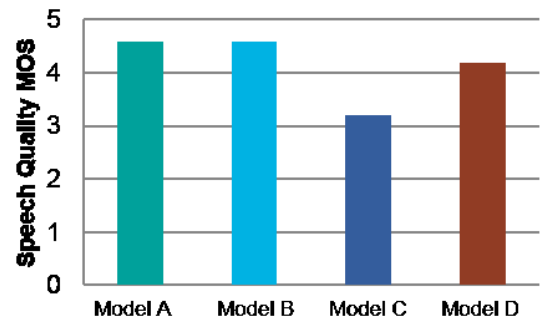
| IP Handover |                         |
|-------------|-------------------------|
| Packet Loss | If % Packet Loss <= 35% |
|             | If % Packet Loss > 35%  |

# Umetrix® Voice

## Evaluate voice quality and experience (Nomad)

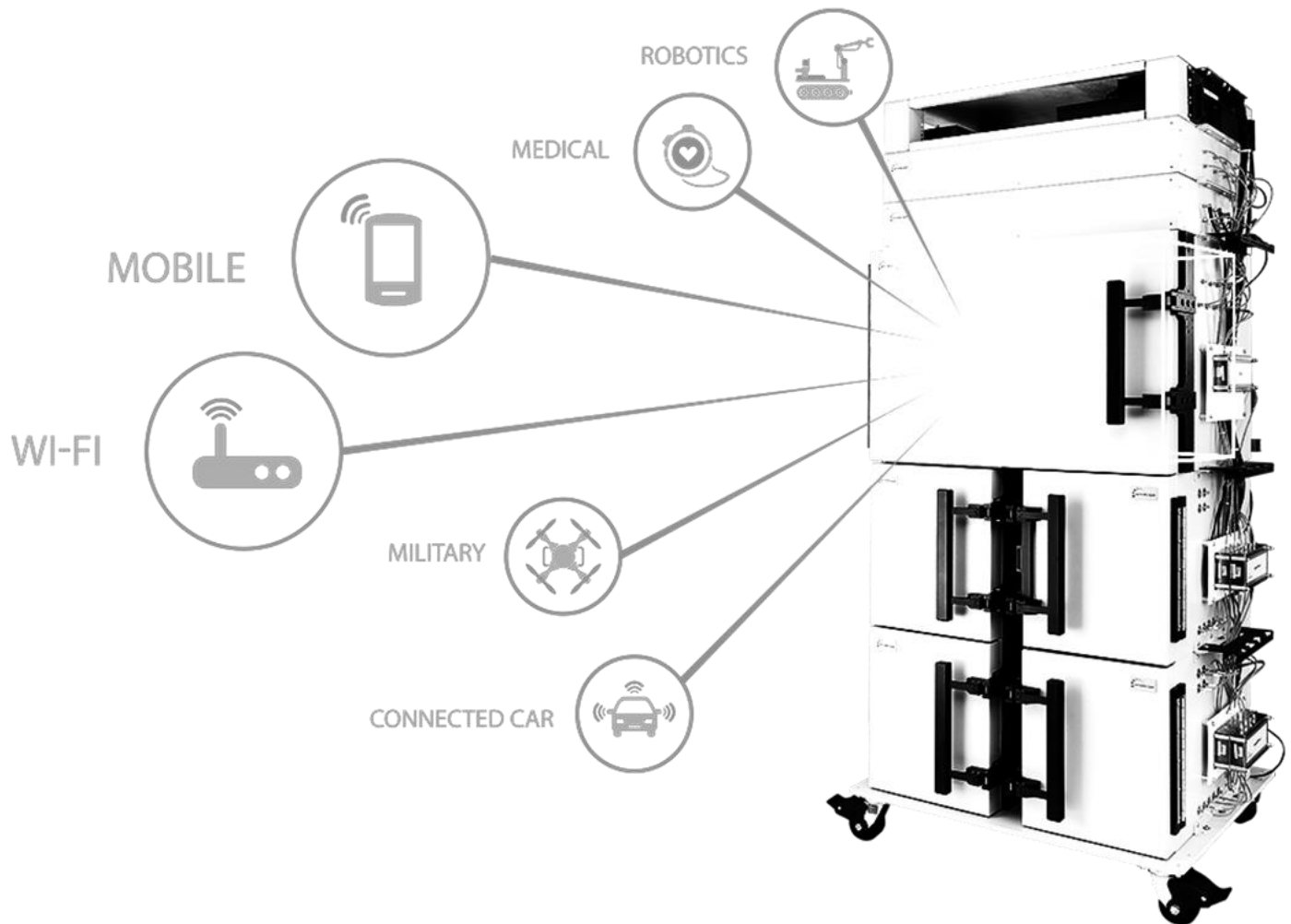
- **Helps launch new voice services** (VoLTE, new EVS codecs, new wideband capabilities, VoWi-Fi, IR94, etc.) and/or benchmark against the competition
- Includes speech quality and call performance
- Quick and easy way to compare various KPIs
- **Identify and resolve problems** with voice services faster
- Ability to compare multiple service areas in a quick manner
- Filtering based on various parameters (RF, coverage, QCI, etc.)
- High-level executive reporting
- Handset/OS agnostic

Consistent metrics on any mobile device



Compare device models, service providers or service types (VoWi-Fi, VoLTE, OTT).

# Spirent's Octobox Solution



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